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INTRODUCTION

1 • 1 HISTORICAL SKETCH

Electromagnetic radiation is one of the most important probes of the structure and dynamics of matter. The absorption of ultraviolet, visible, infrared, and microwave radiation has provided detailed information about electronic, vibrational, and rotational energy levels of molecules and has in some instances enabled the chemist and physicist to determine the structure of complex molecules. Radiofrequency spectroscopy has had an enormous impact on solid-state and molecular physics and physical, inorganic, and organic chemistry. The structure of solids and biological macromolecules has been elucidated by x-ray diffraction experiments. Raman scattering is another spectroscopic technique that provides information similar to that of absorption spectroscopy. When photons impinge on a molecule they can either impart energy to (or gain energy from) the translational, rotational, vibrational, and electronic degrees of freedom of the molecules. They thereby suffer frequency shifts. Thus the frequency spectrum of the scattered light will exhibit resonances at the frequencies corresponding to these transitions. Raman scattering, therefore, provides information about the energy spectra of molecules. This book deals only with the characteristics of the light scattered from translational and rotational degrees of freedom, that is, with what is now commonly called *Rayleigh scattering*.

Recent advances in laser techniques have made possible the measurement of very small frequency shifts in the light scattered from gases, liquids, and solids. Moreover, because of the high intensities of laser sources, it is possible to measure even weakly scattered light. Thus the main difficulties in performing light-scattering experiments encountered in the past are eliminated when lasers are used. This explains the rather remarkable proliferation of laser light-scattering experiments in recent years. The structure and dynamics of such diverse systems as solids, liquid crystals, gels, solutions of biological macromolecules, simple molecular fluids, electrolyte solutions, dispersions of microorganisms, solutions of viruses, membrane vesicles, protoplasm in algae, and colloidal dispersions have now been studied by laser light-scattering techniques.

When light impinges on matter, the electric field of the light induces an oscillating polarization of the electrons in the molecules. The molecules then serve as secondary sources of light and subsequently radiate (scatter) light. The frequency shifts, the angular distribution, the polarization, and the intensity of the scattered light are determined by the size, shape, and molecular interactions in the scattering material. Thus from the light-scattering characteristics of a given system it should be possible, with the aid of electrodynamics and the theory of time-dependent statistical mechanics, to obtain information about the structure and molecular dynamics of the scattering medium.

The basic theory of *Rayleigh scattering* was developed more than a half century ago by Rayleigh, Mie, Smoluchowski, Einstein, and Debye. It is well worth summarizing

some of the high points in the history of the field.

Since the experimental studies by Tyndall (1869) on light scattering from aerosols and the initial theoretical work of Rayleigh (1871, 1881), light scattering has been used to study a variety of physical phenomena. These studies concerned scattering from assemblies of noninteracting particles sufficiently small compared to the wavelength of the light to be regarded as point-dipole oscillators. In his 1881 article Rayleigh also presented an approximate theory for particles of any shape and size having a relative refractive index approximately equal to one.¹ Rayleigh (1899) explained the blue color of the sky and the red sunsets as due to the preferential scattering of blue light by the molecules in the atmosphere. In subsequent papers (Rayleigh 1910, 1914, 1918) Rayleigh derived the full formula for spheres of arbitrary size. For these larger particles there are fixed phase relations between the waves scattered from different points of the same particle, but each scattering element of the particle is regarded as an independent dipole oscillator. Debye (1915) made further contributions to the theory of these large particles and extended the calculations to particles of nonspherical shape.

Gans (1925) also contributed to the theory of large particles of relative refractive index¹ approximately equal to one. The theory of such particles is often referred to as the *Rayleigh-Gans theory*. According to Kerker (1969), "Gans' contribution to this method was hardly significant and it seems more appropriate to call it *Rayleigh-Debye scattering*."

In large particles of relative refractive index much different from one there are not only fixed spatial relations between the scattering elements, there is also a strong dependence of the electric field amplitude on the position in the particle. There are formidable theoretical problems associated with the treatment of these large particles. Only for the case of spheres does there exist a complete solution. Mie (1908), and independently Debye (1909) solved this problem. This type of scattering is now referred to as *Mie scattering*. These problems are discussed at great length in the monographs by Van de Hulst (1957) and by Kerker (1969), and are consequently not considered in this book. Studies of the angular dependence and polarization of the scattered light are now routinely used to study the shapes and sizes of large particles.

Although Rayleigh had developed a theory of light scattering from gases with some success, it was soon found that the intensity of scattering by condensed phases (molecule per molecule) was less than that predicted by his formula by more than one order of magnitude. This effect was correctly attributed to the destructive interference between the wavelets scattered from different molecules, but unfortunately the means of calculating the extent of this interference were not known at that time. Smoluchowski (1908) and Einstein (1910) elegantly circumvented this difficulty by considering the liquid to be a continuous medium in which thermal fluctuations give rise to local inhomogeneities and thereby to density and concentration fluctuations. These authors developed a *fluctuation theory of light scattering*.

According to this theory, the intensity of the scattered light can be calculated from the mean-square fluctuations in density and concentration which in turn can be determined from macroscopic data such as the isothermal compressibility and the concentration-dependence of the osmotic pressure. The intensity of the light is thus obtained without considering the detailed molecular structure of the medium. This phenomenological approach to light scattering has continued to play a very important role in the theory of light scattering, although profound questions regarding the

validity of this approach have been raised [see, for example, Fixman (1955), and more recently Felderhof (1974) and references cited therein].

The scattering from a system of particles whose positions are correlated (governed by a pair-correlation function) was investigated by Zernike and Prins (1927) in connection with the theory of x-ray diffraction of liquids. The same theory applies to light scattering from liquids. This theory was developed by Ornstein and Zernike (1914, 1916, and 1926), who extensively applied it to the study of the intense scattering of light that occurs in the fluid critical region (critical opalescence). The marked increase in the turbidity of the fluids near the gas-liquid critical point is a consequence of the fact that the pair-correlation function in a system near its critical point becomes infinitely long-ranged.

In the foregoing phenomenological theory no attempt was made to describe the effects of molecular optical anisotropy on the intensity, angular dependence, and polarization characteristics, of the scattered light. Subsequent work dealt with a molecular theory of independent optically anisotropic scatterers (Cabannes, 1929; Gans, 1921, 1923). Debye and Zimm and co-workers synthesized the *Rayleigh-Debye* and the phenomenological points of view in the 1940s and developed light scattering as a method for studying molecular weights, sizes, shapes, and interactions of macromolecules in solution. The classic papers on the subject are reprinted in McIntyre and Gornick (1964).

All these studies treated only the intensities of the scattered light. There was, however, a parallel development in light scattering which started with the work of Leon Brillouin (1914, 1922), who predicted a doublet in the frequency distribution of the scattered light due to scattering from thermal sound waves in a solid. This doublet is now known as the *Brillouin doublet*.

In the early 1930s Gross conducted a series of light-scattering experiments on liquids observing the Brillouin doublet and a central or *Rayleigh line* whose peak maximum was unshifted. Landau and Placzek (1934) gave a theoretical explanation of the Rayleigh line using a quasi-thermodynamic approach. They showed that the ratio of the integrated intensity of the central line to that of the doublet is given by the heat-capacity ratio (now known as the *Landau-Placzek ratio*):

$$\frac{I_c}{I_d} = \frac{c_P - c_V}{c_V}$$

This field was carried on by only a few workers, mainly in the Soviet Union and India (see, for example, Fabelinskii, 1968 and references cited therein), but it was not until the development of the laser in the early 1960s that these measurements of frequency changes became a major tool for the study of liquids. The modern hydrodynamic theory of light scattering from liquids is described in Chapters 10, 11, 12, and 13.

With the advent of the laser, another type of experiment became possible. In 1964, Pecora showed that the frequency distribution of light scattered from macromolecular solutions would yield values of the macromolecular diffusion coefficient and under certain conditions might be used to study rotational motion and flexing of macromolecules. These frequency changes are so small that conventional monochromators (or "filters") could not be used to resolve the frequency distribution of the scattered light. In 1964, Cummins, Knable, and Yeh used an *optical-mixing* technique to

spectrally resolve the light scattered from dilute suspensions of polystyrene spheres. Since this pioneering work applications have proliferated, and *optical-mixing spectroscopy* has become a major field of research for workers in chemistry, physics, and biology.

It is the purpose of this book to describe the theory of light-scattering spectroscopy experiments and its applications to major topics of interest to chemists, physicists, and biologists. The older theories concerned with integrated intensities are described in detail only where they are of importance in understanding spectral distribution experiments. The emphasis throughout is on the use of light scattering to study the dynamics of fluctuations in fluids and not on the electrodynamical theory of the interaction of radiation with matter.

1 · 2 SYNOPSIS

In a light-scattering experiment, light from a laser passes through a polarizer to define the polarization of the incident beam and then impinges on the scattering medium. The scattered light then passes through an analyzer which selects a given polarization and finally enters a detector. The position of the detector defines the *scattering angle* θ . In addition, the intersection of the incident beam and the beam intercepted by the detector defines a scattering region of volume V . This is illustrated in Fig. 1.2.1. Pre-laser light-scattering experiments usually used mercury sources. The detector used in

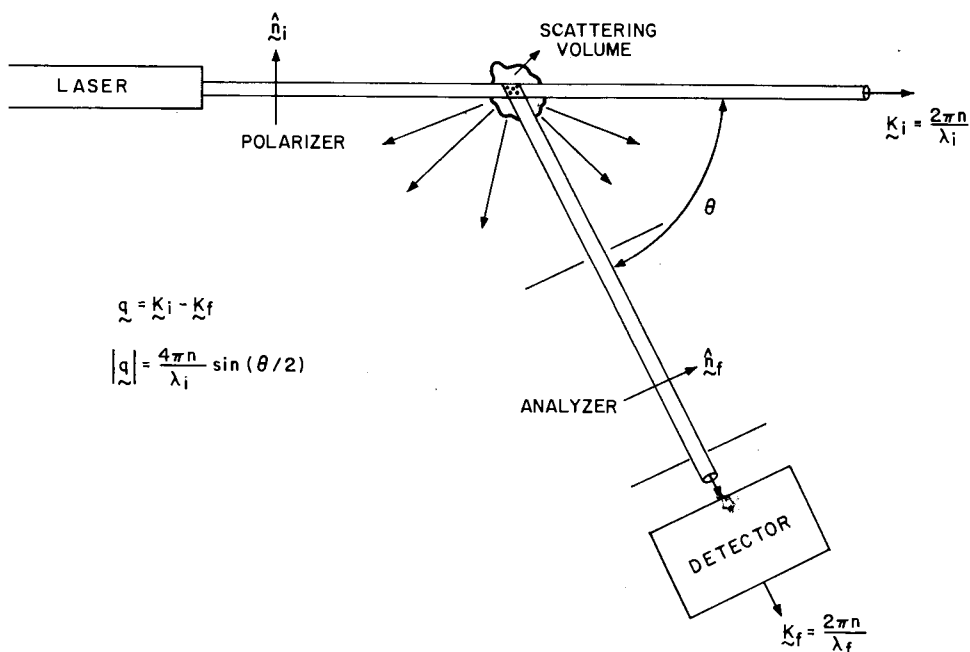


FIG. 1.2.1. A schematic representation of the light-scattering experiment.

these experiments was normally a phototube whose dc output was proportional to the intensity of the scattered light beam. In modern light-scattering experiments the scattered light spectral distribution (or the equivalent) is also measured. In these experiments a photomultiplier is the main detector, but the pre- and postphotomultiplier systems differ depending on the frequency change of the scattered light. The three different methods used, called *filter*, *homodyne*, and *heterodyne methods*, are schematically illustrated in Fig. 1.2.2. Note that homodyne and heterodyne methods use no monochromator or "filter" between the scattering cell and the photomultiplier. These methods are discussed in Chapter 4.

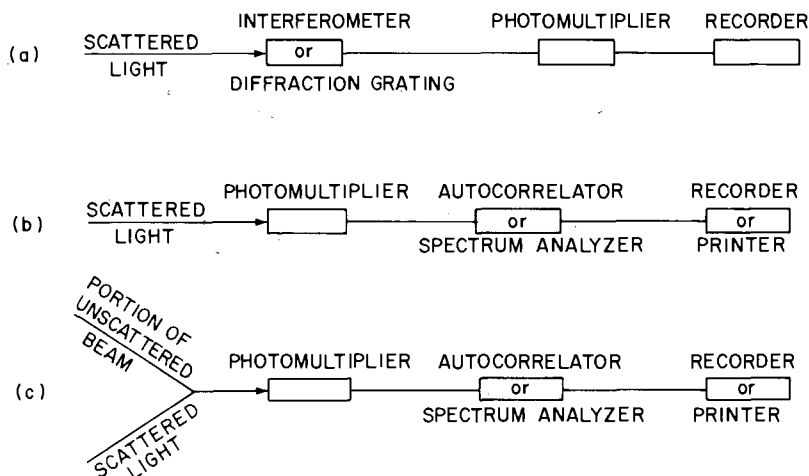


FIG. 1.2.2. Schematic illustration of the various techniques used in light-scattering experiments: (a) filter methods; (b) homodyne; (c) heterodyne.

The spectral characteristics of the scattered light depend on the time scales characterizing the motions of the scatterers. These relationships are discussed in Chapter 3. The quantities measured in light-scattering experiments are the time-correlation function of either the scattered field or the scattered intensity (or their spectral densities). Consequently, time-correlation functions and their spectral densities are central to an understanding of light scattering. They are, therefore, discussed at the outset in Chapter 2.

The theory of light scattering from the simplest systems—dilute solutions or gases composed of spherical molecules—is presented in Chapter 5. This chapter includes discussions of the applications of light scattering to the study of macromolecular diffusion, electrophoretic motions, and the motility of microorganisms. In Chapter 6, a theory of light scattering from a simple model system in chemical equilibrium is presented. Conditions are given under which it might be possible to measure rate constants for chemical reactions by this method, although there have as yet been no unequivocal experimental results that report measurements of rate constants.² An important new technique, fluorescence fluctuation spectroscopy (FFS), is also discussed in this chapter. This technique has been successfully used to measure rate constants for binding of small molecules to macromolecules as well as the diffusion of molecules in membranes. It was thus felt that a treatment of chemical kinetics would be of value to workers in these related areas.

Light scattering can be used to measure rotational time constants for nonspherical molecules in gases and solutions. The theory of scattering from these systems is somewhat more complicated than that from spherical molecules, so that in Chapter 7 several alternative procedures for arriving at some of the results are presented. The mathematical techniques presented in this chapter are useful also for treating related problems such as fluorescence depolarization (Appendix 7.B), electron-spin resonance (ESR), nuclear magnetic relaxation (NMR), and neutron scattering.

When molecules are no longer small compared to the wavelength of light, intramolecular interference becomes important in light-scattering experiments. Since this interference depends on the mass distribution in the molecule, this phenomenon forms the basis for measurements of radii of gyration of macromolecules from integrated intensity measurements. Chapter 8 reviews the theory of light scattering from polymer solutions and also shows how intramolecular interference affects the scattered light frequency dependence and the integrated intensity.

Chapters 9–14 treat systems composed of interacting molecules and the collective modes in these systems. Chapter 9 shows how the long-range Coulomb forces affect light-scattering spectra from solutions. Chapter 10 gives a short treatment of the phenomenological basis of hydrodynamics and then applies it to the calculation of light-scattering spectra. The Brillouin doublet and central line described in Sec. 1.1 as well as the Landau–Placzek ratio are all predicted by this theory.

Chapter 11 reviews the statistical mechanical basis of hydrodynamics and discusses theories that may be used to extend hydrodynamics beyond the “classical” equations discussed in Chapter 10. Chapter 12 applies the statistical mechanical theory to the calculation of depolarized light-scattering spectra from dense liquids where interactions between anisotropic molecules are important.

Chapter 13 includes a short introduction to the theory of nonequilibrium thermodynamics. A discussion of frames of reference in the definition of transport coefficients is given and a systematic theory of diffusion is presented. Fluctuations in electrolyte solutions are analyzed, and the parameters measured in electrophoretic light-scattering experiments are related to conductance and to the transference numbers—quantities usually measured in conventional electrochemistry.

Chapter 14 is devoted to a brief description of collision-induced phenomena and the kinetic theory of gases.

The book concludes with a brief summary of other methods for determining time-correlation functions in Chapter 15.

NOTES

1. By “relative refractive index” is meant the ratio of the refractive index inside a particle to that outside the particle.
2. However, measurement of intramolecular conformational relaxation rates which are discussed in Chapter 8 may in a sense be regarded as a “chemical reaction.”

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LIGHT SCATTERING AND FLUCTUATIONS

2 · 1 INTRODUCTION

In a light-scattering experiment a monochromatic beam of laser light impinges on a sample and is scattered into a detector placed at an angle θ with respect to the transmitted beam (cf. Fig. 1.2.1). The intersection between the incident beam and the scattered beam defines a volume V , called the *scattering volume* or the *illuminated volume*.

In an idealized light-scattering experiment the incident light is a plane electromagnetic wave

$$\mathbf{E}_i(\mathbf{r}, t) = \mathbf{n}_i E_0 \exp i[\mathbf{k}_i \cdot \mathbf{r} - \omega_i t] \quad (2.1.1)$$

of wavelength λ , frequency ω_i , polarization $\mathbf{n}_i$, amplitude E_0 , and wave vector $\mathbf{k}_i$, where $\mathbf{k}_i$ is

$$\mathbf{k}_i = \left(\frac{\omega_i}{c} \right) \hat{\mathbf{k}}_i$$

and $\hat{\mathbf{k}}_i$ is a unit vector specifying the direction of propagation of the incident wave. $\mathbf{E}_i(\mathbf{r}, t)$ is the electric field at the point in space $\mathbf{r}$ at time t . When the molecules in the illuminated volume are subjected to this incident electric field their constituent charges experience a force and are thereby accelerated. According to classical electromagnetic theory, an accelerating charge radiates light. The radiated (or scattered) light field at the detector at a given time is the sum (superposition) of the electric fields radiated from all of the charges in the illuminated volume and consequently depends on the exact positions of the charges.

The molecules in the illuminated region are perpetually translating, rotating, and vibrating by virtue of thermal interactions. Because of this motion the positions of the charges are constantly changing so that the total scattered electric field at the detector will fluctuate in time. Implicit in these fluctuations is important structural and dynamical information about the positions and orientations of the molecules. It is the purpose of this book to show how this structural and dynamical information can be obtained from the fluctuations of the scattered field at the detector.

Thermal molecular motion is erratic, so that the total scattered field varies randomly at the detector. A recording of this field will look very much like a noise pattern. Hence it is no wonder that the theory of noise and fluctuations is relevant to the study of light-scattering spectroscopy. Before deriving the fundamental formulas

of light scattering we present some of the basic ideas in the theory of noise and stochastic processes.

2 · 2 FLUCTUATIONS AND TIME-CORRELATION FUNCTIONS

In light-scattering experiments, the incident light field is sufficiently weak that the system can be assumed to respond linearly to it. The basic theoretical problem is to describe the response of an equilibrium system to this weak incident field, or more precisely, the changes of the light field (frequency shifts, polarization changes, etc.) due to its interaction with the system. This problem has been solved in general for weak probes. The major result of this theory, which is called *linear response theory*, can be simply stated (Zwanzig, 1965). Whenever two systems are weakly coupled to one another (such as radiation weakly coupled to matter), it is only necessary to know how both systems behave in the absence of the coupling in order to describe the way in which one system responds to the other. Furthermore, the response of one system to the other is completely describable in terms of time-correlation functions of dynamical variables.

Time-dependent *correlation functions* have been familiar for a long time in the theory of noise and stochastic processes (Wax, 1954). In recent years they have become very useful in many areas of statistical physics and spectroscopy. Correlation functions provide a concise method for expressing the degree to which two dynamical properties are correlated over a period of time. In this chapter we discuss some of the basic properties of these functions that are relevant to our understanding of light-scattering spectroscopy.

Let us consider a property A that depends on the positions and momenta of all the particles in the system. By virtue of their thermal motions the particles are constantly jostling around so that their positions and momenta are changing in time, and so too is the property A . Although the constituent particles are moving according to Newton's equations (or Schrödinger's equation), their very number makes their motion appear to be somewhat random. The time-dependence of the property $A(t)$ will generally resemble a noise pattern (cf. Fig. 2.2.1).

As an example, consider the pressure on the wall of a cylinder containing a gas in equilibrium. The pressure on the wall at a given time is proportional to the total force on the wall, which in turn is a function of the distances of all the particles from the wall. As the particles move about, the total force fluctuates in time in a very definite manner. The pressure is therefore a fluctuating property. Suppose now that we could couple some kind of gauge to the wall that could respond rapidly to the pressure changes. The needle on this gauge would execute an erratic behavior—it would fluctuate. Since molecular motion is very rapid, the needle would jump around very rapidly. What should be reported as the pressure of the gas? The answer is obvious. The gauge should be read at a large number of time intervals and the results should be averaged. An average over a sufficiently long time (a time long compared with the period of the fluctuation) would yield a fairly reliable pressure. By this we mean that if the same

average were performed at a different time, essentially the same average value would be obtained.

It is clear from this discussion that the measured bulk property of an equilibrium system is simply a time average

$$\bar{A}(t_0, T) = \frac{1}{T} \int_{t_0}^{t_0+T} dt A(t)$$

where t_0 is the time at which the measurement is initiated and T is the time over which it is averaged. The average becomes meaningful only if T is large compared to the period of fluctuation. The ideal experiment would be one in which A is averaged over an infinite time,

$$\bar{A}(t_0) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{t_0}^{t_0+T} dt A(t)$$

It can be shown that under certain general conditions¹ this infinite time average is independent of t_0 . In statistical mechanics it is usually assumed that this is valid. In general a property whose average is independent of t_0 is called a *stationary property* (cf. Section 11.C). In Fig. 2.2.1 we see that the property A fluctuates about this time average, which because of its independence of t_0 can be expressed as

$$\langle A \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt A(t) \quad (2.2.1)$$

The noise signal $A(t)$ in Fig. 2.2.1 displays the following features: the property A at

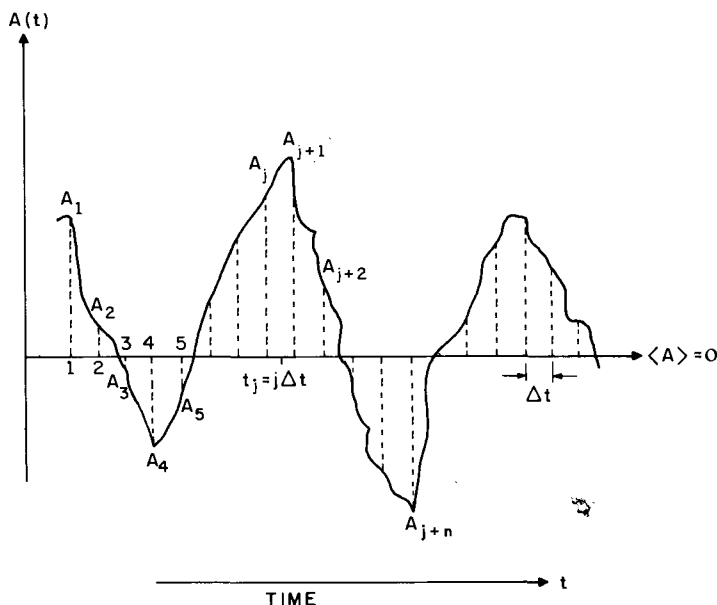


FIG. 2.2.1. The property $A(t)$ fluctuates in time as the molecules move around in the fluid. The time axis is divided into discrete intervals, Δt , and the time average $\langle A \rangle$ is assumed to be zero for convenience.

the two times t , and $t + \tau$ can in general have different values so that $A(t + \tau) \neq A(t)$. Nevertheless when τ is very small compared to times typifying the fluctuations in A , $A(t + \tau)$ will be very close to $A(t)$. As τ increases the deviation of $A(t + \tau)$ from $A(t)$ is more likely to be nonzero. Thus in some sense we can say that the value $A(t + \tau)$ is correlated with $A(t)$ when τ is small but that this correlation is lost as τ becomes large compared with the period of the fluctuations. A measure of this correlation is the autocorrelation function of the property A which is defined by²

$$\langle A(0)A(\tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt A(t)A(t + \tau) \quad (2.2.2)$$

Suppose that the time axis is divided into discrete intervals Δt , such that $t = j\Delta t$; $\tau = n\Delta t$; $T = N\Delta t$ and $t + \tau = (j + n)\Delta t$; and suppose further that the property A varies very little over the time interval Δt . From the definition of the integral it then follows that Eqs. (2.2.1) and (2.2.2) can be approximated by

$$\begin{aligned} \langle A \rangle &\cong \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N A_j \\ \langle A(0)A(\tau) \rangle &\cong \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N A_j A_{j+n} \end{aligned} \quad (2.2.3)$$

where A_j is the value of the property at the beginning of the j^{th} interval. These sums become better approximations to the infinite time averages as $\Delta t \rightarrow 0$.

In optical mixing experiments, a correlator computes time-correlation functions of the scattered field in this discrete manner (see Chapter 4). Of course in any experimental determination the averaging is done over a finite number of steps (finite time).

We introduce the discrete notation in order to clarify the ensuing discussion. What we want to demonstrate is how the time-correlation function varies with time. In Fig. 2.2.1 we present the noise signal $A(t)$. Note that many of the terms in the sum Eq. (2.2.3) are negative. For example, in Fig. 2.2.1 $A_j A_{j+n}$ is negative. Consequently, this sum will involve some cancellation between positive and negative terms. Now consider the case $\langle A(0)A(0) \rangle$. The sum contributing to this is $\sum_j A_j A_j = \sum_j A_j^2$. Since $A_j^2 \geq 0$ all the terms in the sum are positive and we expect the total to be large. What this implies is that³

$$\sum_{j=1}^N A_j^2 \geq \sum_{j=1}^N A_j A_{j+n}$$

or

$$\langle A(0)^2 \rangle \geq \langle A(0)A(\tau) \rangle. \quad (2.2.4)$$

Thus it would appear that the autocorrelation function either remains equal to its initial value for all times τ , in which case A is a constant of the motion (a conserved quantity) or decays from its initial value which is a maximum.

From the foregoing we would expect that the autocorrelation function of a nonconserved, nonperiodic property decays from its initial value $\langle A^2 \rangle$. For times τ large compared to the characteristic time for the fluctuation of A , $A(t)$ and $A(t + \tau)$ are expected to become totally uncorrelated; thus

$$\lim_{\tau \rightarrow \infty} \langle A(0)A(\tau) \rangle = \langle A(0) \rangle \langle A(\tau) \rangle = \langle A \rangle^2 \quad (2.2.5)$$

so that the time-correlation function of a nonperiodic property decays from $\langle A^2 \rangle$ to $\langle A \rangle^2$ in the course of time. This is shown in Fig. 2.2.2.

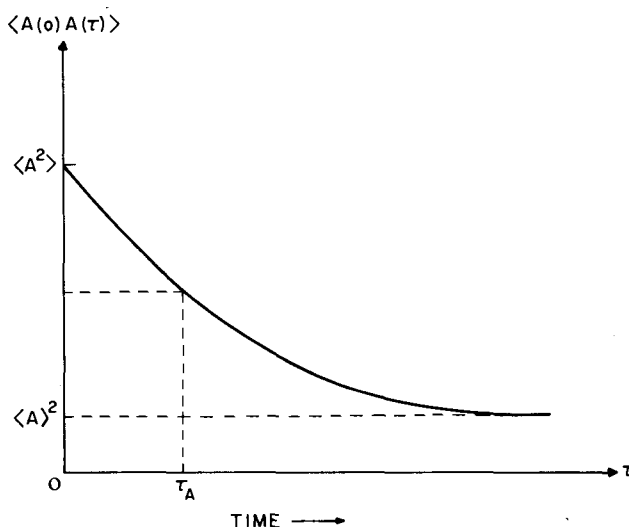


FIG. 2.2.2. The time-correlation function, $\langle A(0)A(\tau) \rangle$. Initially this function is $\langle A^2 \rangle$. For times very long compared to the correlation time, τ_A , the correlation function decays to $\langle A \rangle^2$. [cf. Eq.(2.2.5)].

An actual experimental noise signal is shown in Fig. 2.2.3a. The particular signal is proportional to the intensity of light scattered from a solution of polystyrene spheres of diameter $1.01 \mu\text{m}$. The corresponding (time-averaged) time-correlation function is given in Fig. 2.2.3b. As we show in Chapter 5, the diffusion coefficient of the spheres can be determined from the correlation time of this function. This is discussed in greater detail later.

In many applications the autocorrelation function decays like a single exponential so that

$$\langle A(0)A(\tau) \rangle = \langle A \rangle^2 + \{ \langle A^2 \rangle - \langle A \rangle^2 \} \exp \frac{-\tau}{\tau_r} \quad (2.2.6)$$

where τ_r is called the “relaxation time” or the correlation time of the property. It represents the characteristic decay time of the property. If we define

$$\delta A(t) \equiv A(t) - \langle A \rangle \quad (2.2.7)$$

which is the deviation of the instantaneous value of $A(t)$ from its average value, it is easy to show that⁴

$$\langle \delta A(0)\delta A(\tau) \rangle = \langle A(0)A(\tau) \rangle - \langle A \rangle^2 \quad (2.2.8)$$

and

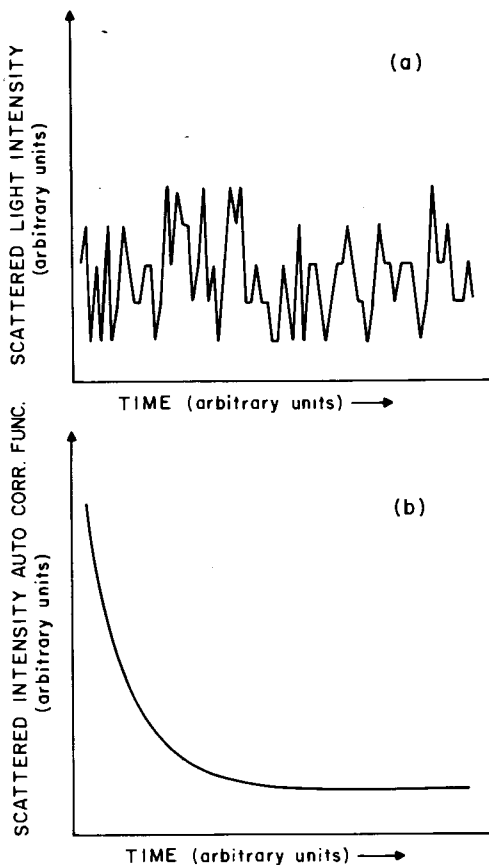


FIG. 2.2.3. (a) Intensity of scattered light (arbitrary units) from an aqueous solution of polystyrene spheres of radius $1.01\mu\text{m}$ as a function of time (arbitrary units). (b) The time-averaged autocorrelation function of the scattered intensity in *a* as a function of time in arbitrary units.

$$\langle \delta A^2 \rangle = \langle \delta A(0) \delta A(0) \rangle = [\langle A^2 \rangle - \langle A \rangle^2] \quad (2.2.9)$$

Combining Eqs. (2.2.6), (2.2.8), and (2.2.9) yields

$$\langle \delta A(0) \delta A(\tau) \rangle = \langle \delta A^2 \rangle \exp \frac{-\tau}{\tau_c} \quad (2.2.10)$$

$\delta A(t)$ is often referred to as a “fluctuation” in that it represents the deviation of the property from its average value. The autocorrelation functions of fluctuations have a simpler structure than the autocorrelation function of the properties themselves because the time invariant part $\langle A \rangle^2$ is removed.

Not all fluctuations decay exponentially. We often want to have some parameter that typifies the time scale for the decay of the correlations. We therefore define the correlation time τ_c to be

$$\tau_c \equiv \int_0^\infty d\tau \frac{\langle \delta A(0) \delta A(\tau) \rangle}{\langle \delta A^2 \rangle} \quad (2.2.11)$$

We note that for exponential decay $\tau_c = \tau_r$. In general, the correlation time will be a some complicated function of all of the relaxation processes contributing to the decay of δA .⁵

Let us summarize; the autocorrelation function is a measure of the similarity between two noise signals $A(t)$ and $A(t + \tau)$. When $\tau = 0$ these two signals are completely in phase with each other and $\langle A(0)A(\tau) \rangle$ is large; as τ increases $A(t)$ and $A(t + \tau)$ get out of phase with each other and the autocorrelation function $\langle A(0)A(\tau) \rangle$ is small.

2 · 3 ENSEMBLE-AVERAGED TIME-CORRELATION FUNCTIONS

The time-correlation functions measured in NMR, neutron, and light-scattering spectroscopy are time averages, whereas in most theoretical calculations what is calculated is the ensemble-averaged time-correlation function. According to Birkhoff's ergodic theorem, these two correlation functions will be identical if the mechanical systems studied are ergodic. Unfortunately, it has not been possible to prove that real systems are ergodic (Uhlenbeck and Ford, 1963). Nevertheless, the predictions of ensemble theory have been so consistent with experiment that throughout the remainder of this book we assume the equivalence between time-averaged and ensemble-averaged time-correlation functions. Hence we devote this section to the definition of the ensemble average both in classical and quantum systems.

The *instantaneous state* of a classical isolated mechanical system of f degrees of freedom is completely specified by f generalized positions ($q_1, \dots, q_f$) and f generalized momenta ($p_1, \dots, p_f$). Such a state can be represented by a point in a $2f$ dimensional cartesian space with orthogonal coordinate axes labeled by these f positions and f momenta.⁶ Given the initial state, $\Gamma_0 \equiv (q_1(0), \dots, p_f(0))$ of the mechanical system the canonical equations of motion,

$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ p_i = -\frac{\partial H}{\partial q_i} \end{cases} \quad i = 1, \dots, f \quad (2.3.1)$$

have unique solutions so that all subsequent states of the system are unambiguously specified for all time. This means that as time goes by the particles move to new positions with new momenta. The specific positions and momenta at any time t , $\Gamma_t \equiv [q_1(t), \dots, p_f(t)]$ are totally determined by the initial state of the system and the canonical equations (or Newton's equations). Of course it is necessary to specify the Hamiltonian of the system, H . For a conservative system it is possible to choose the generalized coordinates and momenta such that

$$H = T(p_1, \dots, p_f) + V(q_1, \dots, q_f) \quad (2.3.2)$$

where T is the kinetic energy and V is the potential energy of the particles. The com-

The behavior of the system can be represented by a trajectory in phase space, as shown in Fig. 2.3.1a.

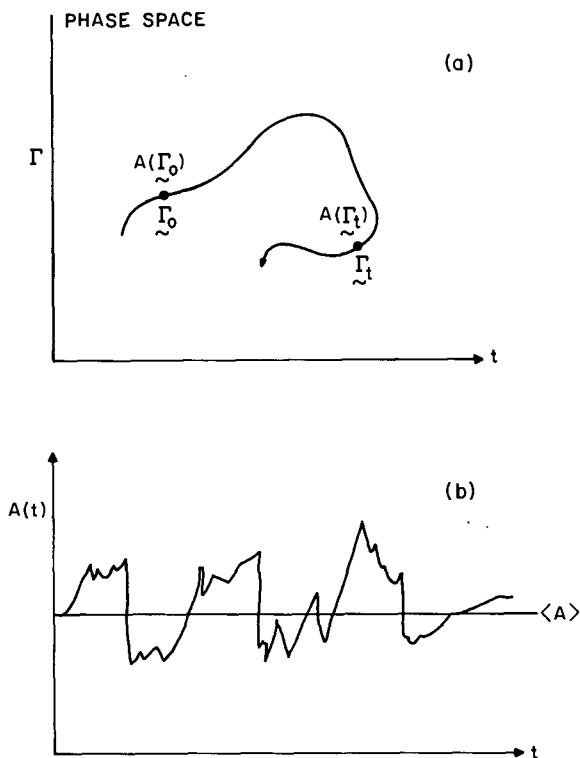


FIG. 2.3.1. (a) Schematic representation of a trajectory in phase space. Γ_0 and Γ_t represent the states of the system at time 0 and t . (b) The variation of the mechanical property A with respect to time corresponding to the trajectory in a .

A mechanical property of a system is a function of the instantaneous state, Γ_t , of the system. For example, if A is a mechanical property, then $A(t) \equiv A(\Gamma_t)$. Examples of mechanical properties are the kinetic energy of a single particle and the number density in the neighborhood of a point in the system. As time goes by a mechanical property will change unless it is a "constant" of the motion. The typical behavior of a mechanical property A corresponding to a given trajectory in phase space is illustrated in Fig. 2.3.1b.

The classical equilibrium ensemble-averaged time-correlation function of the property A is defined as

$$\langle A(0)A(t) \rangle \equiv \int d\Gamma_0 \rho(\Gamma_0) A(\Gamma_0) A(\Gamma_t) \quad (2.3.3a)$$

where the product⁷ $A(\Gamma_0)A(\Gamma_t)$ depends on the time t and on the initial state Γ_0 , and $\rho(\Gamma_0)d\Gamma_0$ is the probability⁸ of finding the system in the initial state Γ_0 .

In quantum statistical mechanics the time-correlation function of the observable A is defined as

$$\langle A(0)A(t) \rangle = \text{Tr} \rho_0 A \exp \frac{i\hat{H}t}{\hbar} A \exp \frac{-i\hat{H}t}{\hbar} = \text{Tr} \rho_0 A(0)A(t) \quad (2.3.3b)$$

where ρ_0 is the equilibrium density matrix, A is the linear Hermitian operator representing the observable A , $\hat{H}$ is the Hamiltonian operator and $A(t) \equiv \exp \frac{i\hat{H}t}{\hbar} A \exp \frac{-i\hat{H}t}{\hbar}$ is the linear operator representing⁹ the observable A at time t . In quantum mechanics a new feature arises. Because $A(t)$ and $A(0)$ do not, in general, commute, $\langle A(0)A(t) \rangle$ and $\langle A(t)A(0) \rangle$ are different functions of the time (in fact, they are complex conjugate pairs). Because most of the applications in this book involve systems that can be treated classically, we shall not dwell on this point here.

One particular time-correlation function of the many that we have occasion to study in this book is

$$F(\mathbf{q}, t) = \langle \rho_{-\mathbf{q}}(0) \rho_{\mathbf{q}}(t) \rangle \quad (2.3.4)$$

where the property $\rho_{\mathbf{q}}(t)$ is

$$\rho_{\mathbf{q}}(t) = \sum_{j=1}^N \exp i\mathbf{q} \cdot \mathbf{r}_j(t) \quad (2.3.5)$$

where $\mathbf{q}$ is an arbitrary vector and $\mathbf{r}_j(t)$ is the center-of-mass position of the j^{th} molecule in the system. We note that $\rho_{\mathbf{q}}(t)$ is the Fourier transform¹⁰ with respect to $\mathbf{r}$ of the property

$$\rho(\mathbf{r}, t) = \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) \quad (2.3.6)$$

where $\delta(x)$ is the Dirac delta function. This property is the microscopic number density at the point $\mathbf{r}$. This is readily verified since the delta function contributes 1 to any average only if $\mathbf{r}_j(t)$ is in the neighborhood of the point $\mathbf{r}$ and zero otherwise. Thus the sum in Eq. (2.3.6) counts the number of particles in the neighborhood of the point $\mathbf{r}$ at time t , and thereby the number density. $F(\mathbf{q}, t)$ is a measure of the correlation between spatial Fourier components of the number density at two different times. This function can be determined by light-scattering experiments and, as we shall see in subsequent chapters, contains a great deal of information about static and dynamic properties of fluid systems.

2 • 4 THE SPECTRAL DENSITY



The spectral density (or power spectrum) $I_A(\omega)$ of a time-correlation function $\langle A^*(0)A(t) \rangle$ is defined as¹¹

$$I_A(\omega) \equiv \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt e^{-i\omega t} \langle A^*(0)A(t) \rangle \quad (2.4.1)$$

where A^* is the complex conjugate of A .

This quantity plays an important role in much of what follows. In fact, as we shall see, what is sometimes measured in light scattering is the spectral density of the electric field of the scattered light. Let us dwell for a moment on some properties of these functions. Fourier inversion of Eq. (2.4.1) leads to an expression for the time-correlation function in terms of the spectral density.

$$\langle A^*(0)A(t) \rangle = \int_{-\infty}^{+\infty} d\omega e^{i\omega t} I_A(\omega) \quad (2.4.2)$$

Thus $\langle A(0)A(t) \rangle$ and $I_A(\omega)$ are Fourier transforms of one another and an experimental determination of one as a function of its arguments is sufficient for the determination of the other. In fact, it is $I_A(\omega)$ rather than the time-correlation function that is often measured in an experiment. Moreover, it should be noted that the equilibrium mean-square value of the property A is found by setting $t = 0$ in the preceding formula so that

$$\langle |A|^2 \rangle = \langle |A(0)|^2 \rangle = \int_{-\infty}^{+\infty} d\omega I_A(\omega) \quad (2.4.3)$$

Thus $I_A(\omega)d\omega$ can be interpreted as the "amount" of $|A|^2$ in the frequency interval $(\omega, \omega + d\omega)$. This interpretation can be clarified as follows. Figure 2.4.1 schematically

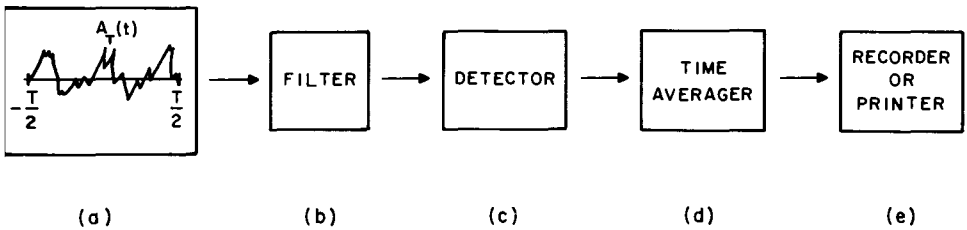


FIG. 2.4.1. Schematic of apparatus for measuring the spectral density of a fluctuating variable.

illustrates the main features of an apparatus for measuring $I_A(\omega)$. The signal $A_T(t)$, measured for a period T , is first passed through a filter which allows through only a very narrow range of frequencies. The signal that gets through the filter, $A_{TO}(t)$, then impinges on a detector whose output is proportional to $|A_{TO}(t)|^2$. The output of the detector is then averaged over the time interval T for which the signal is defined and the result, $\langle |A_{TO}|^2 \rangle_T$ is then suitably recorded.

A function $A(t)$ measured over a time interval $(-T/2, T/2)$ can always be expressed in terms of its Fourier components so that

$$A_T(t) = \frac{1}{\sqrt{T}} \sum_n A_n \exp i\omega_n t \quad (2.4.4)$$

where $\omega_n = \frac{2\pi}{T}n$ and $\{A_n\}$ are the Fourier coefficients.

On passing through the filter, some of the frequency components of A_T are filtered out. We can describe the action of the filter by a set of numbers F_n which we allow to be zero if the frequency ω_n is not allowed through and to be one otherwise. Thus the signal $A_T(t)$ is transformed by the filter to

$$A_{To}(t) = \frac{1}{\sqrt{T}} \sum_n F_n A_n \exp i\omega_n t \quad (2.4.5)$$

The output of the detector is proportional to the square of this function

$$|A_{To}(t)|^2 = \frac{1}{T} \sum_{nn'} F_{n'} F_n A_{n'}^* A_n \exp i(\omega_n - \omega_{n'}) t$$

The averager then performs the time average,

$$\begin{aligned} \langle |A_{To}|^2 \rangle &= \frac{1}{T} \int_{-T/2}^{T/2} dt |A_{To}(t)|^2 \\ &= \frac{1}{T} \int_{-T/2}^{T/2} \frac{dt}{T} \sum_{nn'} F_{n'} F_n A_{n'}^* A_n \exp i(\omega_n - \omega_{n'}) t \end{aligned}$$

where Eq. (2.4.5) has been substituted. Performing this integration over t we obtain¹²

$$\langle |A_{To}|^2 \rangle = \frac{1}{T} \sum_n F_n^2 |A_n|^2 \quad (2.4.6)$$

This can be related to the time-correlation function¹³

$$\langle A^*(t) A(t + \tau) \rangle_T = \frac{1}{T} \int_{-T/2}^{T/2} dt A_T^*(t) A_T(t + \tau) \quad (2.4.7)$$

When the Fourier series expansion, Eq. (2.4.4), is substituted into this, we obtain

$$\langle A^*(t) A(t + \tau) \rangle_T = \sum_{n,n'} \frac{A_n'^* A_n}{T} \int_{-T/2}^{T/2} \frac{dt}{T} \exp i(\omega_n - \omega_{n'}) t \exp i\omega_n \tau$$

But this is simply equal to¹⁴

$$\langle A^*(t) A(t + \tau) \rangle_T = \sum_n \frac{|A_n|^2}{T} \exp i\omega_n \tau \quad (2.4.8)$$

Multiplication by $\exp -i\omega_m \tau$ and integration over τ from $-T/2$ to $T/2$ then gives

$$|A_m|^2 = \int_{-T/2}^{T/2} d\tau \langle A^*(t) A(t + \tau) \rangle_T \exp -i\omega_m \tau = 2\pi I_A^T(\omega_m) \quad (2.4.9)$$

where $I_A^T(\omega_m)$ is by definition the spectral density of the time-correlation function $\langle A^*(t) A(t + \tau) \rangle_T$. Substituting Eq. (2.4.9) into Eq. (2.4.6) we now obtain

$$\langle |A_{To}|^2 \rangle_T = \frac{2\pi}{T} \sum_n I_A^T(\omega_n) F_n^2 \quad (2.4.10)$$

This can be written in another form. The separation between adjacent frequencies $\Delta\omega_n = \omega_{n+1} - \omega_n = 2\pi/T$, so that

$$\langle |A_{To}|^2 \rangle_T = \sum_n \Delta\omega_n I_A^T(\omega_n) F_n^2$$

If $T \rightarrow \infty$, $\Delta\omega_n \rightarrow 0$, the above sum can be written as an integral

$$\lim_{T \rightarrow \infty} \langle |A_{TO}|^2 \rangle_T = \int_{-\infty}^{+\infty} d\omega I_A(\omega) |F(\omega)|^2 \quad (2.4.11)$$

where $I_A(\omega)$ is the spectral density of A determined by an average over an infinite time [i.e., $I_A(\omega) = \lim_{T \rightarrow \infty} I_A^T(\omega)$] and $|F(\omega)|^2$ is a filter function. If the filter is a “narrow-band filter

$$|F(\omega)|^2 = \begin{cases} 1 & \omega_0 \leq \omega \leq \omega_0 + \Delta\omega \\ 0 & \text{otherwise} \end{cases} \quad (2.4.12)$$

then

$$\lim_{T \rightarrow \infty} \langle |A_{TO}|^2 \rangle_T = I_A(\omega_0) \Delta\omega \quad (2.4.13)$$

Thus from a time-average of $|A_{TO}|^2$ we can obtain the spectral density $I_A(\omega_0)$, and by tuning the filter through different values of ω_0 we can determine the spectrum of the fluctuation A . From its definition we see that $\lim_{T \rightarrow \infty} \langle |A_{TO}|^2 \rangle_T$, and correspondingly $I_A(\omega_0)\Delta\omega$ can be interpreted as the “amount” of $|A_T|^2$ passed through the filter, that is, the amount in the frequency interval $(\omega_0, \omega_0 + \Delta\omega)$, thus clarifying our assertion following Eq. (2.4.3).

The foregoing result is quite general. If $E_T(t)$ is the electric field of a scattered light wave, if the filter is an interferometer, grating or prism (all of which are narrow-band filters), and if the detector is a photomultiplier (all photomultipliers are square-law detectors), then according to Eq. (2.4.13) the output is

$$\lim_{T \rightarrow \infty} \langle |E_{TO}|^2 \rangle_T = I_E(\omega_0) \Delta\omega \quad (2.4.14)$$

where $I_E(\omega_0)$ is the spectral density of the electric field autocorrelation function and ω_0 is defined by the filter. This experimental situation corresponds to the usual optical spectrometer (see Chapter 4 for further details) which consequently measures the spectral density $I_E(\omega)$. We conclude that if the filter is tuned through all frequencies, the spectral density

$$I_E(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\tau \langle E^*(t) E(t + \tau) \rangle \exp i\omega\tau \quad (2.4.15)$$

can be determined as a function of frequency, and the time-correlation function can be determined by Fourier inversion. This means that the property $I_E(\omega)$ which is measured in a filter experiment and the correlation function $\langle E^*(0)E(t) \rangle$ measured in heterodyne experiments are Fourier-transform pairs.¹⁵

The term $I_E(\omega)$ can be measured directly either by filter experiments or by heterodyne methods using a spectrum analyzer instead of a correlator. As we have noted before, the technique of choice depends upon the time scale of the fluctuations.

It is clear from Eq. (2.4.15) and Section 1.2 that the light-scattering spectrum is determined from autocorrelation functions of the electric field at the detector. Thus the goal of any theory of light scattering is to show how important physical properties of the scattering medium can be extracted from the measured time-correlation functions.

NOTES

1. Birkhoff proved this for an ergodic system; that is, a system that is metrically indecomposable (Uhlenbeck and Ford, 1963).
2. The correlation between two different properties A and B is similarly given by the *cross-correlation functions*

$$\langle A(0)B(\tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt A(t)B(t + \tau)$$

and

$$\langle B(0)A(\tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt B(t)A(t + \tau)$$

3. This inequality can be proved using Schwartz's inequality, according to which

$$\left| \sum_j A_j B_j \right|^2 \leq \left| \sum_j A_j^2 \right| \left| \sum_j B_j^2 \right|$$

If we take $B_j = A_{j+n}$, divide both sides by N^2 , take the limit $N \rightarrow \infty$ and then recognize that:

$$(a) \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N A_j^2 = \langle A^2 \rangle$$

$$(b) \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N A_{j+n}^2 = \langle A^2 \rangle$$

$$(c) \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N A_j A_{j+n} = \langle A(0)A(\tau) \rangle$$

it follows that

$$|\langle A(0)A(\tau) \rangle|^2 \leq \langle A^2 \rangle^2$$

Since $\langle A(0)A(\tau) \rangle$ is real, the inequality

$$\langle A(0)A(\tau) \rangle \leq \langle A^2 \rangle$$

follows. The limit in (b) follows from the presumed independence of the time average from the time at which averaging is initiated.

It should also be noted that the correlation function can under some circumstances be periodic.

4. Substitute $A(t) = \langle A \rangle + \delta A(t)$ into Eq. (2.2.2). Then note that

$$\langle A(0)A(\tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt \{ \langle A \rangle^2 + \langle A \rangle [\delta A(t) + \delta A(t + \tau)] + \delta A(t) \delta A(t + \tau) \}$$

Since $\langle A \rangle$ is a constant and $\langle \delta A(t) \rangle = 0 = \langle \delta A(t + \tau) \rangle$;

$$\langle A(0)A(\tau) \rangle = \langle A \rangle^2 + \langle \delta A(0) \delta A(\tau) \rangle$$

5. This definition of $|\tau_c|$ is not always useful. There are cases in which, for instance, the correlation function decays in such a way that $|\tau_c|$ as defined by Eq. (2.2.11) is zero. For example, the correlation function may have positive and negative regions that cancel. In such cases Eq. (2.2.11) does not provide an adequate measure of the decay time.
6. This space is called *phase space*.
7. In many cases the property A may have complex values. In such cases we define the autocorrelation function as

$$\langle A^*(0)A(\tau) \rangle = \int d\Gamma_0 \rho(\Gamma_0) A^*(\Gamma_0) A(\Gamma_0 + \tau)$$

where A^* is the complex conjugate of A .

8. $\rho(\Gamma_0)$ can be any of the equilibrium ensemble distribution functions. For example, in the canonical ensemble $\rho(\Gamma_0) = Q^{-1} \exp[-\beta H(\Gamma_0)]$ where Q is the canonical partition function, $Q = \int d\Gamma_0 e^{-\beta H}$.
9. This is found in the Heisenberg representation.
10. The Fourier transform of a function $f(\mathbf{r})$ is denoted $f_{\mathbf{q}}$ where

$$f_{\mathbf{q}} \equiv \int d^3r e^{i\mathbf{q} \cdot \mathbf{r}} f(\mathbf{r})$$

11. More precisely, the spectral density is defined as

$$I_A(\omega) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt e^{-i\omega t} e^{-\varepsilon|t|} \langle A^*(0)A(t) \rangle$$

where the factor $e^{-\varepsilon|t|}$ ($\varepsilon > 0$) is introduced to ensure the existence of the integral. $I_A(\omega)$ is the time Fourier transform of $\langle A^*(0)A(t) \rangle$. Equations (2.4.1) and (2.4.2) are Fourier-transform pairs.

12. Where we use the fact that

$$\int_{-T/2}^{T/2} dt \exp i(\omega_n - \omega_{n'})t = T\delta_{n,n'}$$

13. This is the time-correlation function averaged over a finite time T .

14. See Note 12.

15. The average *intensity* I of light passing through the filter is $\frac{c}{8\pi} \langle |E_{T0}|^2 \rangle$, where c is the speed of light. It follows that $I(\omega)d\omega$, the intensity of scattered light with frequency between ω and $\omega + d\omega$ is related to $I_E(\omega)$ by

$$I(\omega) = \frac{c}{8\pi} I_E(\omega)$$

that is by a proportionality constant $c/8\pi$. Since we are not interested in absolute intensities, we will always refer to $I_E(\omega)$ as the scattered-intensity spectrum.

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BASIC LIGHT SCATTERING THEORY

3 · 1 INTRODUCTION

The theory of light scattering can be developed on the basis of quantum field theory. The major results of this theory differ little (and in most cases not at all) from a classical theory of light scattering. Since the major emphasis in this book is on what the light scattering spectrum tells us about physical systems, we do not dwell on the electrodynamical theory. It suffices for our purposes to discuss only the elementary theory, keeping to a minimum the mathematical derivations, which are relegated to Appendix 3.A.

In the classical theory of light scattering an incident electromagnetic field exerts a force on the charges in the scattering volume. These accelerating charges then radiate light. The incident field is said to polarize the medium. When visible light is incident upon the medium the atoms in a subregion of the illuminated volume, small compared to the cube of the incident light wavelength, see essentially the same incident electric field. If many subregions of equal size are considered, the scattered electric field is the superposition of the scattered fields from each of them. If the subregions are optically identical, that is, each has the same dielectric constant, there will be no scattered light in other than the forward direction. This is so because the wavelets scattered from each subregion are identical except for a phase factor that depends on the relative positions of the subregions. If we ignore surface effects it is clear that for a large medium, each subregion can always be paired with another subregion whose scattered field is identical in amplitude but opposite in phase and will thus cancel, leaving no net scattered light in other than the forward direction. If, however, the regions are optically different, that is, have different dielectric constants, then the amplitudes of the light scattered from the different subregions are no longer identical. Complete cancellation will no longer take place, and there will be scattered light in other than the forward direction. Thus in this semimacroscopic view, originally introduced by Einstein, light scattering is a result of local fluctuations in the dielectric constant of the medium (Einstein, 1910). It is easy to understand how such fluctuations may take place. We know from kinetic theory that molecules are constantly translating and rotating so that the instantaneous dielectric constant of a given subregion (which depends on the positions and orientations of the molecules) will fluctuate and thus give rise to light scattering.

3 · 2 RESULTS FROM ELECTROMAGNETIC THEORY

Consider a nonmagnetic, nonconducting, nonabsorbing medium¹ with average dielectric constant ϵ_0 (and refractive index $n = \sqrt{\epsilon_0}$). Let the incident electric field be a plane wave of the form

$$\mathbf{E}_i(\mathbf{r}, t) = \mathbf{n}_i E_0 \exp i(\mathbf{k}_i \cdot \mathbf{r} - \omega_i t) \quad (3.2.1)$$

where $\mathbf{n}_i$ is a unit vector in the direction of the incident electric field; E_0 is the field amplitude; $\mathbf{k}_i$ is the propagation vector² (or wave vector) and ω_i is the angular frequency.

This plane wave is incident upon a medium that has a *local* dielectric constant

$$\epsilon(\mathbf{r}, t) = \epsilon_0 \mathbf{I} + \delta\epsilon(\mathbf{r}, t) \quad (3.2.2)$$

where $\delta\epsilon(\mathbf{r}, t)$ is the dielectric constant fluctuation tensor at position $\mathbf{r}$ and time t and $\mathbf{I}$ is the second-rank unit tensor.

It may be shown by the methods given in Appendix 3.A that the component of the scattered electric field at a large distance R from the scattering volume with polarization $\mathbf{n}_f$, propagation vector $\mathbf{k}_f$, and frequency ω_f is³

$$E_s(R, t) = \frac{E_0}{4\pi R \epsilon_0} \exp i k_f R \int_V d^3r \exp i(\mathbf{q} \cdot \mathbf{r} - i\omega_f t) [\mathbf{n}_f \cdot [\mathbf{k}_f \times (\mathbf{k}_f \times (\delta\epsilon(\mathbf{r}, t) \cdot \mathbf{n}_i)]] \quad (3.2.3)$$

where the subscript V indicates that the integral is over the scattering volume (see Fig. 3.2.1). The vector $\mathbf{q}$ is defined in terms of the scattering geometry as

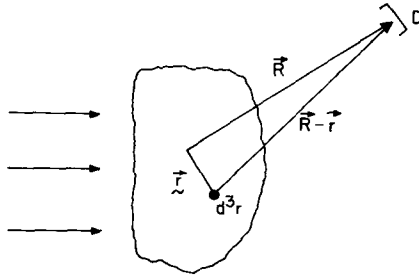
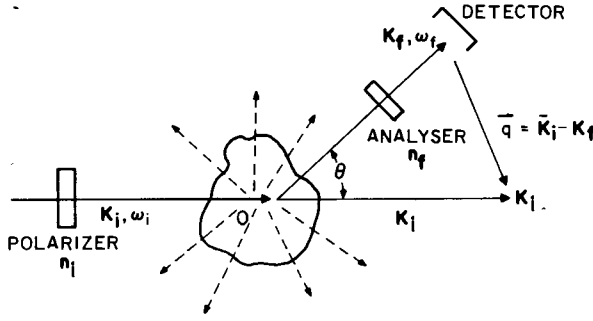


FIG. 3.2.1. The total radiated field at the detector is the superposition of the fields radiated from all infinitesimal volumes d^3r at positions $\mathbf{r}$ with respect to the center of the illuminated volume V . The detector is at position $\mathbf{R}$ with respect to the center of the illuminated volume.

$$\mathbf{q} = \mathbf{k}_i - \mathbf{k}_f \quad (3.2.4)$$

where $\mathbf{k}_i$ and $\mathbf{k}_f$ point, respectively, in the directions of propagation of the incident wave and the wave that reaches the detector. The angle between $\mathbf{k}_i$ and $\mathbf{k}_f$ is called the *scattering angle* θ . This is illustrated in Fig. 3.2.2. The magnitudes of $\mathbf{k}_i$ and $\mathbf{k}_f$ are respectively $2\pi n/\lambda_i$ and $2\pi n/\lambda_f$, where λ_i and λ_f are the wave lengths in vacuo of the incident and scattered radiation and n is the refractive index of the scattering medium. It



$$q^2 = k_i^2 + k_f^2 - 2\mathbf{k}_i \cdot \mathbf{k}_f \approx 4k_i^2(1 - \cos \theta) = 4k_i^2 \sin^2 \theta/2$$

FIG. 3.2.2. Light of polarization $\mathbf{n}_i$ and wave vector $\mathbf{k}_i$ is scattered in all directions. Only scattered light of wave vector $\mathbf{k}_f$ and polarization $\mathbf{n}_f$ arrives at the detector. The scattering vector $\mathbf{q} = \mathbf{k}_i - \mathbf{k}_f$ is defined by the geometry. Since the scattered wave has essentially the same wavelength as the incident wave, $k_f \simeq (2\pi n)/\lambda_i = k_i$, it follows from the law of cosines that $q = 2k_i \sin \theta/2$.

is usually the case that the wavelength of the incident light is changed very little in the scattering process so that

$$|\mathbf{k}_i| \cong |\mathbf{k}_f|$$

Thus the triangle in Fig. 3.2.1 is an isosceles triangle and the magnitude of $\mathbf{q}$ can be found from the law of cosines,

$$q^2 = |\mathbf{k}_f - \mathbf{k}_i|^2 = k_f^2 + k_i^2 - 2\mathbf{k}_i \cdot \mathbf{k}_f = 2k_i^2 - 2k_i^2 \cos \theta = 4k_i^2 \sin^2 \frac{\theta}{2}$$

$$q = 2k_i \sin \frac{\theta}{2} = \frac{4\pi n}{\lambda_i} \sin \frac{\theta}{2} \quad (3.2.5)$$

This is the Bragg condition. It specifies the wave vector component of the dielectric constant fluctuation that will give rise to scattering at an angle θ .

Equation (3.2.3) can be expressed in terms of the spatial Fourier transform of the dielectric fluctuation

$$\delta \epsilon(\mathbf{q}, t) = \int_V d^3r \exp i\mathbf{q} \cdot \mathbf{r} \delta \epsilon(\mathbf{r}, t) \quad (3.2.6)$$

as

$$E_s(R, t) = \frac{E_0}{4\pi R \epsilon_0} \exp i(k_f R - \omega_i t) \{ \mathbf{n}_f \cdot [\mathbf{k}_f \times \mathbf{k}_f \times (\delta \epsilon(\mathbf{q}, t) \cdot \mathbf{n}_i)] \} \quad (3.2.7)$$

Equation (3.2.7) can be simplified by working out the vector cross products.⁴ Then

$$E_s(R, t) = \frac{-k_f^2 E_0}{4\pi R \epsilon_0} \exp i(k_f R - \omega_i t) \delta \epsilon_{if}(\mathbf{q}, t) \quad (3.2.8)$$

where

$$\delta \epsilon_{if}(\mathbf{q}, t) \equiv \mathbf{n}_f \cdot \delta \epsilon(\mathbf{q}, t) \cdot \mathbf{n}_i \quad (3.2.9)$$

is the component of the dielectric constant fluctuation tensor along the initial and final polarization directions. The time-correlation function (cf. Sections 2.2 and 2.3) of E_s can be evaluated from Eq. (3.2.8)

$$\langle E_s^*(R, 0) E_s(R, t) \rangle = \frac{k_f^4 |E_0|^2}{16\pi^2 R^2 \epsilon_0^2} \langle \delta\epsilon_{if}(\mathbf{q}, 0) \delta\epsilon_{if}(\mathbf{q}, t) \rangle \exp -i\omega t \quad (3.2.10)$$

The spectral density of light scattered into the detector such that

$$(\mathbf{n}_i, \mathbf{k}_i, \omega_i) \rightarrow (\mathbf{n}_f, \mathbf{k}_f, \omega_f)$$

can be determined by substitution of Eq. (3.2.10) into Eq. (2.4.15)

$$I_{if}(\mathbf{q}, \omega_f, R) = \left[\frac{I_0 k_f^4}{16\pi^2 R^2 \epsilon_0^2} \right] \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt \langle \delta\epsilon_{if}(\mathbf{q}, 0) \delta\epsilon_{if}(\mathbf{q}, t) \rangle \exp i(\omega_f - \omega_i)t \quad (3.2.11)$$

where $I_0 \equiv |E_0|^2$

Note the following in Eq. (3.2.11): (a) $I_{if} \propto \lambda^{-4}$ (b) $I_{if} \propto R^{-2}$, and (c) I_{if} depends on ω_i and ω_f only through their difference

$$\omega \equiv \omega_i - \omega_f \quad (3.2.12)$$

which is the frequency change in the scattering process. The inverse λ^4 dependence in (a) indicates, for instance, that blue light is scattered more than red light. This results in the blue colors of the sky and oceans. It also indicates that, other things being equal, radio waves would not be scattered as much as visible light. As a consequence of the larger scattering intensities, it is much easier to do scattering experiments with visible light than with the longer wavelength infrared or radio waves. The R^{-2} dependence is just the attenuation expected for a spherical wave. The frequency change occurs only if $\delta\epsilon(\mathbf{q}, t)$ varies with time,⁵ that is, scattering could occur from "frozen" fluctuations but the frequency of the scattered wave would be identical to that of the incident wave.

In Eq. (3.2.11) $I_{if}(\mathbf{q}, \omega_f, R)$ is thus seen to be proportional to the spectral density of the dielectric constant fluctuations, $I_{if}^s(\mathbf{q}, \omega)$

$$I_{if}^s(\mathbf{q}, \omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt e^{-i\omega t} \langle \delta\epsilon_{if}^*(\mathbf{q}, 0) \delta\epsilon_{if}(\mathbf{q}, t) \rangle \quad (3.2.13)$$

where the proportionality constant is

$$A = \frac{k_f^4 I_0}{16\pi^2 R^2 \epsilon_0^2} \quad (3.2.14)$$

In the following chapters the autocorrelation function of the dielectric constant fluctuations is denoted

$$I_{if}^s(\mathbf{q}, t) = \langle \delta\epsilon_{if}^*(\mathbf{q}, 0) \delta\epsilon_{if}(\mathbf{q}, t) \rangle \quad (3.2.15)$$

Thus we see that the scattering event that produces the wave vector change $\mathbf{q}$ and fre-

quency shift ω is due entirely to dielectric constant fluctuations of wave vector $\mathbf{q}$ and frequency ω

The scattering event may be viewed in terms of momentum and energy conservation. A "photon" suffers an energy change from $\hbar\omega_i$ to $\hbar\omega_f$ and a momentum change from $\hbar\mathbf{k}_i$ to $\hbar\mathbf{k}_f$, thereby creating (or annihilating) an excitation in the scattering medium of energy $\hbar\omega$ and momentum $\hbar\mathbf{q}$, where from energy and momentum conservation⁶

$$\hbar\omega = \hbar\omega_i - \hbar\omega_f$$

$$\hbar\mathbf{q} = \hbar\mathbf{k}_i - \hbar\mathbf{k}_f$$

In order to calculate a light-scattering spectrum we must have a model for the mechanism by which dielectric fluctuations decay. The remainder of this book is devoted primarily to the study of these fluctuations.

What is sometimes⁷ measured in light-scattering spectroscopy is $I_{if}^e(\mathbf{q}, \omega)$ at a given scattering angle (that is for fixed $\mathbf{q}$) as a function of ω . The spectrum can be quite complicated, and in fact contains a great deal of interesting information. Only with highly monochromatic sources can the spectrum be sufficiently resolved for these experiments to be useful. Prior to the advent of the laser, only the total intensity $I_{if}^e(\mathbf{q})$ was measured routinely. This is equivalent to integrating the spectrum $I_{if}^e(\mathbf{q}, \omega)$ over all frequencies, that is, to finding the area under the spectrum. Even this integrated intensity provides important information about the system, as we see later. Note that the integral of Eq. (3.2.13) over frequency is⁸

$$I_{if}^e(\mathbf{q}) = \int_{-\infty}^{+\infty} d\omega I_{if}^e(\mathbf{q}, \omega) = \int_{-\infty}^{+\infty} dt \delta(t) \langle \delta\epsilon_{if}^*(\mathbf{q}, 0) \delta\epsilon_{if}(\mathbf{q}, t) \rangle$$

so that

$$I_{if}^e(\mathbf{q}) = \langle |\delta\epsilon_{if}(\mathbf{q})|^2 \rangle \quad (3.2.16)$$

Integrated intensities therefore provide information about the mean-square fluctuations of ϵ for given wave vectors.

3 · 3 MOLECULAR APPROACH TO LIGHT SCATTERING

Equation (3.2.13) is an expression for the scattered light spectral density in terms of dielectric constant fluctuations. Nowhere in this treatment was it necessary to determine the explicit dependence of these fluctuations on molecular properties. In fact, this theoretical expression is purely phenomenological. Any attempt to write this formula in molecular terms will necessarily involve some degree of approximation. Nevertheless, a molecular formulation will contribute much to our intuitive understanding of light scattering and will also be useful for practical application.

Let us first consider what happens when an incident monochromatic beam [c.f. Eq. (3.2.1)] impinges on a single molecule which has an anisotropic polarizability specified by a polarizability tensor α . The incident light induces a dipole moment

$$\dot{\boldsymbol{\mu}}(t) = \boldsymbol{\alpha} \cdot \mathbf{E}(t) \quad (3.3.1)$$

which varies with time. According to classical radiation theory a time varying dipole emits electromagnetic radiation. The electric field of the radiated or scattered light (cf. Jackson, 1965) at the detector is proportional to $\hat{\mathbf{k}}_f \times (\hat{\mathbf{k}}_f \times \ddot{\boldsymbol{\mu}}(t'))$ where t' is the retarded time (cf. Appendix 3. A) and $\ddot{\boldsymbol{\mu}}$ is the second time derivative of $\boldsymbol{\mu}$. Following the same kind of argument given in Appendix 3.A and evaluating the vector cross products, it is simple to show that the scattered field at the detector with polarization $\mathbf{n}_f$ is proportional to $\alpha_{if}(t) e^{i\mathbf{q} \cdot \mathbf{r}(t)}$ where

$$\alpha_{if}(t) = \mathbf{n}_f \cdot \boldsymbol{\alpha}(t) \cdot \mathbf{n}_i \quad (3.3.2)$$

is the component of the molecular polarizability tensor along $\mathbf{n}_i$ and $\mathbf{n}_f$; $\mathbf{r}(t)$ is the position of the center of mass of the molecule at time t and $\mathbf{q}$ is the usual scattering vector⁹ given in Eq. (3.2.4). $\alpha_{if}(t)$ varies in time because the molecule rotates and vibrates, while the phase factor, $e^{i\mathbf{q} \cdot \mathbf{r}(t)}$, varies in time because the molecule translates.

In a fluid, if the molecules are electronically weakly coupled, that is, if the electronic states of the molecules are not perturbed very much by their neighbors, it is reasonable to assume that the light scattered from the assembly of molecules in the illuminated volume will be a superposition of amplitudes scattered from each of the molecules; that is, the scattered field will be proportional to a sum of terms

$$\sum_j' \alpha_{if}^j(t) \exp i\mathbf{q} \cdot \mathbf{r}_f(t)$$

where the index stands for the j th molecule in the assembly (and the prime on the sum indicates that the sum is only over molecules in the illuminated volume). This leads to an interference pattern that is modulated by molecular motions. The spectral density of the scattered field will thus be proportional to¹⁰

$$I_{if}^g(\mathbf{q}, \omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt e^{-i\omega t} I_{if}^g(\mathbf{q}, t) \quad (3.3.3a)$$

where

$$I_{if}^g(\mathbf{q}, t) = \langle \delta\alpha_{if}^*(\mathbf{q}, 0) \delta\alpha_{if}(\mathbf{q}, t) \rangle \quad (3.3.3b)$$

and

$$\delta\alpha_{if}(\mathbf{q}, t) = \sum_{j=1}^N \alpha_{if}^j(t) \exp i\mathbf{q} \cdot \mathbf{r}(t) \quad (3.3.4)$$

is obviously the spatial Fourier component of the polarizability density (cf. Section 2.3).

$$\delta\alpha_{if}(\mathbf{r}, t) = \sum_{j=1}^N \alpha_{if}^j(t) \delta(\mathbf{r} - \mathbf{r}_f(t)) \quad (3.3.5)$$

This ‘molecular’ theory is clearly an approximation. In general, when two molecules collide they suffer distortions in their electronic charge distributions. These distortions persist for the time it takes a molecule to cross the effective range of the in-

termolecular potential. This is of order 10^{-13} sec. Thus we expect on the basis of physical intuition that Eq. (3.3.3) will not account for short-time phenomena—what is now called *collision-induced light scattering* (see Chapter 14). These short-time effects lead to very broad bands with exponential wings in the spectrum (Gelbart, 1974), on top of which sit relatively narrow bands relating to the slower motions. These latter bands can be predicted on the basis of Eq. (3.3.3) and therefore lend credence to this approximation. In general, α_{if}^j is not the isolated molecule polarizability, but is some renormalized or effective polarizability.

The polarizability component $\alpha_{if}^j(t)$ can be regarded as the sum of two terms: (a) the polarizability of the molecule frozen in its equilibrium nuclear configuration and (b) the term linear in the vibrational displacements. Equation (3.3.3) will then consist of four terms (a) a term only involving the rigid molecule polarizability, (b) two cross-terms that are linear in the vibrational displacements, and (c) a term quadratic in the vibrational displacements. The cross terms (b) usually average to zero whereas the term (c) gives rise to the vibration-rotation Raman spectrum. This can be seen as follows. The term (c) contains the vibrational displacements Q in products such as $Q'(0)Q(t)$, where Q and Q' belong to different molecules but refer to the same vibrational mode. First, we note that the displacements on different molecules are usually weakly coupled so that the only significant terms derive from autocorrelations of Q on the same molecule. Second, we note that in a normal mode $Q(t) \sim \cos \Omega t$ where Ω is the frequency of the mode. Thus these Raman terms give rise to shifts of frequency of the bands by Ω and $-\Omega$. The former is the anti-Stokes band and the latter, the Stokes band of the Raman spectrum. The dependence of α_{if}^j on the vibrational displacements therefore gives rise to several bands corresponding to the different "Raman active" normal modes. A mode is Raman allowed if $\left(\frac{\partial \alpha_{if}^j}{\partial Q}\right)_0$ is unequal to zero. This is the well-known classical theory of the Raman effect. It incorrectly predicts Stokes and anti-Stokes lines of equal intensity. The quantum theory proceeds in a similar way but correctly accounts for the observed differences in intensity of these bands (cf. Placzek, 1934). The frequency displacements of these vibration-rotation Raman bands are usually in the range 100 to 4000 cm^{-1} . These frequency changes are much greater than those usually observed for terms of type (a) which only involve rotations of the molecules through the dependence of $\alpha_{if}^j(t)$ on the instantaneous molecular orientation and translations through the dependence of the phase factors $\exp i\mathbf{q} \cdot \mathbf{r}_j(t)$ on the positions of the molecules.

In this book we are concerned only with that part of the spectrum dependent on pure rotations and translations of molecules. No further discussion of the vibrational Raman scattering is given. Thus Eq. (3.3.3) is used, but always with the rigid-frame polarizabilities. We refer to this type of scattering as "*Rayleigh-Brillouin*" scattering.¹¹

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3 • 4 SCATTERING GEOMETRIES

The scattering expressions in Sections 3.2 and 3.3 have been written in general tensor notation and hence are independent of any specific laboratory coordinate system used for a scattering experiment. It is, however, convenient for many applications to use

specific scattering geometries. In this book we use two basic scattering geometries throughout.¹² We describe below these geometries and write out the components of the dielectric constant fluctuation tensor in each.

The plane defined by the initial and final wave vectors of the light is called the *scattering plane*. It is necessary to define the scattering geometry in relation to the scattering plane. The two geometries used are indicated in Figs. 3.4.1 and 3.4.2. For convenience scattering geometry I will be used in connection with the macroscopic theories of light scattering and scattering geometry II, in connection with the molecular theories.

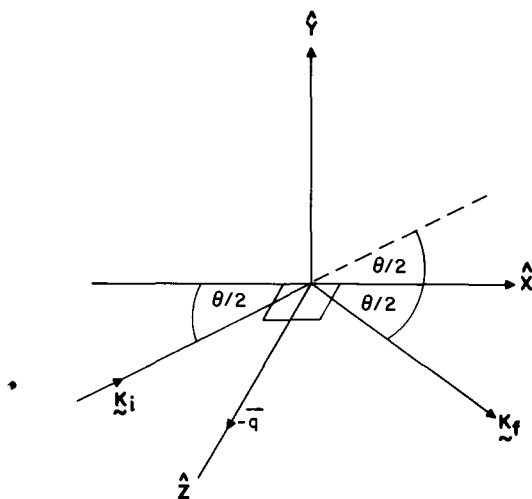


FIG. 3.4.1. *Scattering geometry I.* In this geometry the XZ plane is the scattering plane. The angle $(\mathbf{k}_i, \mathbf{k}_f)$ is the scattering angle, and the scattering vector $\mathbf{q} = \mathbf{k}_i - \mathbf{k}_f$ is antiparallel to the Z axis.

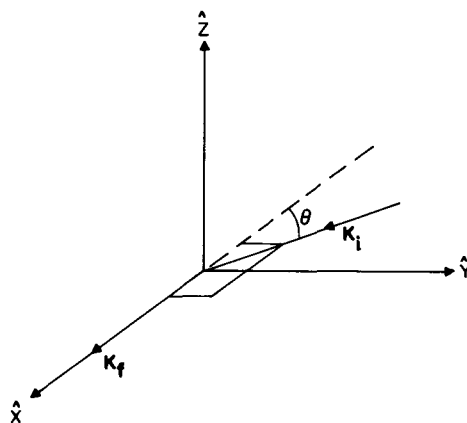


FIG. 3.4.2. *Scattering geometry II.* In this geometry the XY plane is the scattering plane, θ is the scattering angle, and $\mathbf{q} = \mathbf{k}_i - \mathbf{k}_f$ does not lie along any labeled axis.

Four different polarization directions are defined in Fig. 3.4.3. The specific components of the dielectric fluctuations or polarizability fluctuations that are responsible for each of these spectral components are:

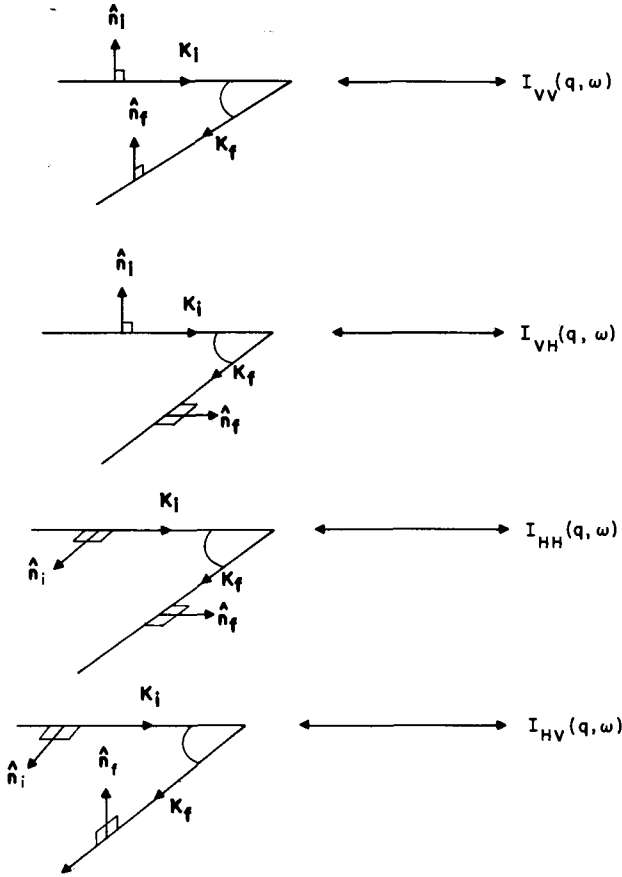


FIG. 3.4.3. Four different pairs of polarization directions commonly used in light-scattering experiments.

Geometry I: $\mathbf{q} = -q\hat{z}$

$$\delta\epsilon_{VV}(\mathbf{q}, t) = \delta\epsilon_{YY}(\mathbf{q}, t)$$

$$\delta\epsilon_{VH}(\mathbf{q}, t) = \delta\epsilon_{YX}(\mathbf{q}, t) \sin \frac{\theta}{2} - \delta\epsilon_{YZ}(\mathbf{q}, t) \cos \frac{\theta}{2}$$

$$\delta\epsilon_{HV}(\mathbf{q}, t) = \delta\epsilon_{XY}(\mathbf{q}, t) \sin \frac{\theta}{2} + \delta\epsilon_{ZY}(\mathbf{q}, t) \cos \frac{\theta}{2} \quad (3.4.1)$$

$$\begin{aligned} \delta\epsilon_{HH}(\mathbf{q}, t) &= \delta\epsilon_{XX}(\mathbf{q}, t) \sin^2 \frac{\theta}{2} - \delta\epsilon_{ZZ}(\mathbf{q}, t) \cos^2 \frac{\theta}{2} \\ &\quad + [\delta\epsilon_{ZX}(\mathbf{q}, t) - \delta\epsilon_{XZ}(\mathbf{q}, t)] \sin \frac{\theta}{2} \cos \frac{\theta}{2} \end{aligned}$$

Geometry II: $\mathbf{q} = q[\hat{x}(1 + \cos \theta) - \hat{y} \sin \theta]$

$$\delta\epsilon_{VV}(\mathbf{q}, t) = \delta\epsilon_{ZZ}(\mathbf{q}, t)$$

$$\delta\epsilon_{VH}(\mathbf{q}, t) = \delta\epsilon_{ZY}(\mathbf{q}, t)$$

$$\delta\epsilon_{HV}(\mathbf{q}, t) = \delta\epsilon_{XZ}(\mathbf{q}, t) \sin \theta + \delta\epsilon_{YZ}(\mathbf{q}, t) \cos \theta \quad (3.4.2)$$

$$\delta\epsilon_{HH}(\mathbf{q}, t) = \delta\epsilon_{XY}(\mathbf{q}, t) \sin \theta + \delta\epsilon_{YY}(\mathbf{q}, t) \cos \theta$$

The subscripts V and H correspond to directions that are vertical and horizontal with respect to the scattering plane. I_{VV} is sometimes called the *polarized* component and I_{VH} and I_{HV} are usually called the *depolarized* components. Usually $I_{VH} = I_{HV}$. Systems exhibiting optical activity are a common exception to this rule¹³. I_{HH} is often a linear combination of I_{VV} and I_{VH} .

APPENDIX 3.A DERIVATION OF THE SCATTERED FIELD

The Maxwell equations for a nonconducting, nonmagnetic medium may be used to obtain the basic equation for the scattered field Eq. (3.2.4). We follow the treatment of Landau and Lifshitz (1960). For more background on the Maxwell equations the interested reader should consult this text.

For simplicity we consider at the outset a medium with a local dielectric constant tensor

$$\boldsymbol{\epsilon} = \epsilon_0 \mathbf{I} + \delta\boldsymbol{\epsilon} \quad (3.A.1)$$

and note that in general the scattered field is much lower in amplitude than the incident field.

If the incident plane wave fields are $\mathbf{E}_i$, $\mathbf{D}_i$, $\mathbf{H}_i$ and scattered fields are $\mathbf{E}_s$, $\mathbf{D}_s$, and $\mathbf{H}_s$.¹⁴ Then the totals of these fields at a point in the scattering medium are

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_i + \mathbf{E}_s \\ \mathbf{D} &= \mathbf{D}_i + \mathbf{D}_s \\ \mathbf{H} &= \mathbf{H}_i + \mathbf{H}_s \end{aligned} \quad (3.A.2)$$

Since $(\mathbf{E}, \mathbf{D}, \mathbf{H})$ and $(\mathbf{E}_i, \mathbf{D}_i, \mathbf{H}_i)$ each satisfy the Maxwell equations it is easy to show that the scattered fields $\mathbf{E}_s$, $\mathbf{D}_s$, and $\mathbf{H}_s$ also obey the Maxwell Equations. Thus

$$\nabla \times \mathbf{E}_s = - \frac{1}{c} \frac{\partial \mathbf{H}_s}{\partial t} \quad (3.A.3)$$

$$\nabla \times \mathbf{H}_s = \frac{1}{c} \frac{\partial \mathbf{D}_s}{\partial t} \quad (3.A.4)$$

$$\nabla \cdot \mathbf{H}_s = 0 \quad (3.A.5)$$

$$\nabla \cdot \mathbf{D}_s = 0 \quad (3.A.6)$$

$\mathbf{H}_s$ may be eliminated from these equations by taking the curl of Eq. (3.A.3) and substituting Eq. (3.A.4) into it;

$$\nabla \times \nabla \times \mathbf{E}_s = \frac{-1}{c^2} \frac{\partial^2 \mathbf{D}_s}{\partial t^2} \quad (3.A.7)$$

The total displacement vector $\mathbf{D}$ and the total electric field vector $\mathbf{E}$ are related through the dielectric constant (Eq. 3.A.1),

$$\begin{aligned}\mathbf{D} &= (\epsilon_0 \mathbf{I} + \delta \boldsymbol{\epsilon}) \cdot (\mathbf{E}_i + \mathbf{E}_s) \\ &= \epsilon_0 \mathbf{E}_i + (\delta \boldsymbol{\epsilon}) \cdot \mathbf{E}_i + \epsilon_0 \mathbf{E}_s + (\delta \boldsymbol{\epsilon}) \cdot \mathbf{E}_s\end{aligned}\quad (3.A.8)$$

From Eq. (3.A.2) and the fact that $\mathbf{D}_i = \epsilon_0 \mathbf{E}_i$, Eq. (3.A.8) becomes

$$\mathbf{D}_s = \epsilon_0 \mathbf{E}_s + (\delta \boldsymbol{\epsilon}) \cdot \mathbf{E}_i \quad (3.A.9)$$

In Eq. (3.A.7) we have neglected the second-order term $(\delta \boldsymbol{\epsilon}) \cdot \mathbf{E}_s$.

Then, solving Eq. (3.A.9) for $\mathbf{E}_s$, substituting the result into Eq. (3.A.7) and using Eq. (3.A.9), we obtain¹⁵ an inhomogeneous wave equation

$$\nabla^2 \mathbf{D}_s - \left(\frac{\epsilon_0}{c^2} \right) \frac{\partial^2 \mathbf{D}_s}{\partial t^2} = -\nabla \times \nabla \times (\delta \boldsymbol{\epsilon} \cdot \mathbf{E}_i) \quad (3.A.10)$$

Equation (3.A.10) may be simplified by defining a new vector, $\boldsymbol{\pi}$ (the Hertz vector) by

$$\mathbf{D}_s = \nabla \times \nabla \times \boldsymbol{\pi} \quad (3.A.11)$$

Substituting (3.A.11) into (3.A.10), we see that the Hertz vector satisfies a wave equation, with a simple source term $-(\delta \boldsymbol{\epsilon}) \cdot \mathbf{E}_i$,

$$\nabla^2 \boldsymbol{\pi} - \left(\frac{\epsilon_0}{c^2} \right) \frac{\partial^2 \boldsymbol{\pi}}{\partial t^2} = -(\delta \boldsymbol{\epsilon}) \cdot (\mathbf{E}_i) \quad (3.A.12)$$

The formal solution of Eq. (3.A.12) is

$$\boldsymbol{\pi}(\mathbf{R}, t) = \frac{1}{4\pi} \int d^3r \frac{\delta \boldsymbol{\epsilon}(\mathbf{r}, t')}{|\mathbf{R} - \mathbf{r}|} \cdot \mathbf{E}_i(\mathbf{r}, t') \quad (3.A.13)$$

where $\mathbf{R}$ and $\mathbf{r}$ are defined in Fig. 3.A.1 and t' is the retarded time

$$t' = t - \frac{\sqrt{\epsilon_0}}{c} |\mathbf{R} - \mathbf{r}| \quad (3.A.14)$$

If Eq. (3.2.1) for $\mathbf{E}_i$ is substituted into Eq. (3.A.13), the operations in Eq. (3.A.11) are performed upon the result to obtain $\mathbf{D}_s$ and, it is noted that at the detector (assumed to be immersed in a medium of dielectric constant ϵ_0) $\mathbf{D}_s = \epsilon_0 \mathbf{E}_s$, we obtain

$$\mathbf{E}_s(\mathbf{R}, t) = \nabla \times \nabla \times \left[\frac{E_0}{4\pi\epsilon_0} \int_j d^3r \frac{1}{|\mathbf{R} - \mathbf{r}|} [\delta \boldsymbol{\epsilon}(\mathbf{r}, t') \cdot \mathbf{n}_i] \exp i(\mathbf{k}_i \cdot \mathbf{r} - \omega_i t') \right] \quad (3.A.15)$$

Since the detector is a large distance from the scattering medium, $|\mathbf{R} - \mathbf{r}|$ may be expanded in a power series

$$|\mathbf{R} - \mathbf{r}| \cong \mathbf{R} - \mathbf{r} \cdot \hat{\mathbf{k}}_f + \dots \quad (3.A.16)$$

where $\hat{\mathbf{k}}_f$ is a unit vector in the direction of $\mathbf{R}$. From Eq. (3.A.14) it follows that

$$t' \cong t - \frac{\sqrt{\epsilon_0}}{c} (R - \mathbf{r} \cdot \hat{\mathbf{k}}_f) \quad (3.A.17)$$

To proceed, we perform a Fourier analysis of $\delta\mathbf{e}(\mathbf{r}, t')$ over an interval T which allows us to write

$$\delta\mathbf{e}(\mathbf{r}, t') = \sum_p \delta\mathbf{e}_p(\mathbf{r}) \exp i\Omega_p t' \quad (3.A.18)$$

where $\Omega_p = (2\pi/T)p$. The only frequency components Ω_p that contribute to this sum are those typifying the natural rotational and translational motions of the system which are typically smaller than 10^{13}sec^{-1} and which are always small compared with the incident light frequency ω_i ; that is, $\omega_i \gg \Omega_p$ for all relevant Ω_p . Substituting Eqs. (3.A.17) and (3.A.18) into Eq. (3.A.15) and defining

$$\begin{cases} \omega_f \equiv \omega_i - \Omega_p \\ \mathbf{k}_p \equiv \frac{\sqrt{\epsilon_0}}{c} \omega_f \hat{\mathbf{k}}_f \\ \mathbf{q}_p \equiv \mathbf{k}_i - \mathbf{k}_p \end{cases} \quad (3.A.19)$$

we find

$$E_s(R, t) = \frac{E_0}{4\pi\epsilon_0 R} \sum_p \exp i[k_p R - \omega_i t] \mathbf{k}_p \times \left[\mathbf{k}_p \times \int_V d^3r \exp i(\mathbf{k}_i - \mathbf{k}_p \hat{\mathbf{k}}_f) \cdot \mathbf{r} \right. \\ \left. \delta\mathbf{e}_p(\mathbf{r}) (\exp i\Omega_p t) \cdot \mathbf{n}_i \right] \quad (3.A.20)$$

where we have ignored terms of higher order than $(1/R)$. Now we note that because $\Omega_p \ll \omega_i$, k_p is to a very good approximation

$$k_p \simeq \frac{\sqrt{\epsilon_0}}{c} \omega_i = k_i \cong k_f \quad (3.A.21)$$

where the second equality follows from $\omega_i = (ck_i/n)$ and where the refractive index is $n = \sqrt{\epsilon_0}$. It follows that $k_p \hat{\mathbf{k}}_f \simeq k_i \hat{\mathbf{k}}_f$. Thus to this order of approximation $\delta\mathbf{e}_p$ and $\exp i\Omega_p t$ are the only p -dependent quantities in the sum, so that Eq. (3.A.20) becomes

$$E_s(R, t) = \frac{E_0}{4\pi\epsilon_0 R} \exp i[k_f R - \omega_i t] \mathbf{k}_f \times \left[\mathbf{k}_f \times \int_V d^3r (\exp i\mathbf{q} \cdot \mathbf{r}) (\delta\mathbf{e}(\mathbf{r}, t) \cdot \mathbf{n}_i) \right] \quad (3.A.22)$$

where we have substituted Eq. (3.A.18) in the form $\delta\mathbf{e}(\mathbf{r}, t) = \sum_p \delta\mathbf{e}_p(\mathbf{r}) \exp i\Omega_p t$, and where we have defined

$$\begin{aligned} \mathbf{k}_f &= k_i \hat{\mathbf{k}}_f \\ \mathbf{q} &= \mathbf{k}_i - \mathbf{k}_f \end{aligned} \quad (3.A.23)$$

It should be noted that $k_f = k_i$ and that $\mathbf{q}$ and $\mathbf{k}_f$ so defined are consistent with the

drawing in Fig. (3.2.1) and with Eq. (3.2.5). Note that Eq. (3.A.22) is free of the retarded time and depends only on the real time t .

If we take the component of $\mathbf{E}_s$ in the direction $\mathbf{n}_f$ and set $\varepsilon_0 = 1$, we obtain

$$E_s(R, t) = \frac{E_0}{4\pi R} \exp i[k_f R - \omega_f t] \int_V d^3r e^{i\mathbf{q} \cdot \mathbf{r}} \mathbf{n}_f \cdot [\mathbf{k}_f \times \mathbf{k}_f \times [\delta \mathbf{e}(\mathbf{r}, t) \cdot \mathbf{n}_i]] \quad (3.A.24)$$

which is the same as Eq. (3.2.3). Equation (3.2.13) may then be obtained from this by the simple operations indicated in the text.

It should be pointed out that this result and all the other results given in this book refer only to the single scattering of an incident light wave. Multiple scattering is entirely neglected because we have omitted higher order terms in $\delta \mathbf{e}$ after Eq. (3.A.10).

NOTES

1. Many of the applications of light scattering are to ionic solutions, which are conducting media. However since the ions are massive the free charge density will vary on a much slower time scale than that specified by the laser frequency ($\sim 10^{14}$ Hz). Thus the medium may be considered to be nonconducting as far as this derivation is concerned. Also, since we consider the medium to be nonabsorbing, we are restricted to incident wave lengths which are not resonant with any molecular transitions of the scattering medium. The reader interested in resonant light scattering should consult the appropriate references (Peticolas, 1972, Bauer et al., 1975).
2. The propagation vector is a vector with a direction parallel to the direction of propagation of the wave and magnitude $|\mathbf{k}_i| = 2\pi n/\lambda_i$ where λ_i is the wavelength in vacuo.
3. This formula ignores both multiple scattering and "local field" effects (e.g., see Gelbart, 1974).
4. where we use the identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

and have used the property that $\mathbf{n}_f \cdot \hat{\mathbf{k}}_f = 0$; that is, the final polarization $\mathbf{n}_f$ must be perpendicular to the scattered field vector $\hat{\mathbf{k}}_f$.

5. This may be easily seen from Eq. (3.2.11). If $\langle \delta \varepsilon_{if}(0) \delta \varepsilon_{if}(t) \rangle$ is independent of time, the time integral is reduced to a delta function $\delta(\omega_f - \omega_i)$, and there is no frequency shift. Actually there would be a small shift due to the recoil of the whole system required by momentum conservation. This is a high-order effect that is completely neglected in our treatment of light scattering.
6. This language is appropriate to the quantum theory of light scattering.
7. In the filter technique or heterodyne technique with spectrum analyzer (see Chapter 4).
8. Where we have used the integral representation of the delta function

$$1/2\pi \int_{-\infty}^{+\infty} d\omega e^{i\omega t} = \delta(t)$$

9. In the molecular theory it is a subtle problem to show that the correct q which enters the phase factors involves the medium wave vector with the refractive index and not the vacuum wave vectors (e.g., see Felderhoff, 1974).
10. This same result can be derived from the quantum mechanical T matrix formulation of the scattering process (e.g., see Goldberger and Watson, 1964). Also, the integrated intensity is proportional to

$$I_{if}^{\alpha}(\mathbf{q}, 0) = \langle \delta \alpha_{if}^*(\mathbf{q}, 0) \delta \alpha_{if}(\mathbf{q}, 0) \rangle$$

From macroscopic electrodynamics it can be shown that $\varepsilon = 1 + 4\pi\alpha$ where α is the polarizability. It follows that $\delta \varepsilon = 4\pi\delta\alpha$ and consequently that

$$I_{if}^{\varepsilon}(\mathbf{q}, t) = 16\pi^2 I_{if}^{\alpha}(\mathbf{q}, t)$$

11. There is some arbitrariness in what we call Raman and Rayleigh-Brillouin scattering. It should be

noted that terms of type (a) above also include what is sometimes called "pure rotational" Raman scattering. It would, in fact, be reasonable to think of Rayleigh-Brillouin scattering as translation-rotation Raman scattering.

12. For some scattering problems it is useful to consider less restrictive scattering geometries than those used here.
13. For a discussion of the conditions under which this "reciprocity" relation is true see Perrin (1942).
14. $\mathbf{E}$, $\mathbf{D}$, and $\mathbf{H}$ are, respectively, the electric, dielectric displacement, and magnetic fields.
15. The vector identity

$$\nabla \times \nabla \times \mathbf{A} = -\nabla^2 \mathbf{A} + \nabla(\nabla \cdot \mathbf{A})$$

has also been used.

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THE LIGHT-SCATTERING EXPERIMENT

4 · 1 INTRODUCTION

It is not the purpose of this chapter to give details or evaluations of the different experimental methods, but only to outline some of the main techniques presently employed so that the different methods of data presentation can be related to some of the theoretical correlation functions described in this book. The reader interested in further details should consult the review articles by Cummins and Swinney (1970), Fleury and Boon (1974), and Cummins and Pike (1974) and the references contained therein.

In a light-scattering experiment a beam of light is focused onto a region of a fluid and is scattered into a detector. Polarizers and analyzers are used to define the polarizations of the incident and scattered light beams, respectively. Physically, the instantaneous scattered field can be regarded as the superposition of waves scattered from the individual scattering centers. This scattered field therefore fluctuates in response to the molecular motions of the scatterers. A variety of detection schemes are used to analyze the time-dependence of these fluctuations. The detection method used in a particular experiment depends on the time scale of these fluctuations. *Filter methods* are used to study relatively rapid molecular dynamic processes, that is, those that occur on a time scale faster than about 10^{-6} sec. *Optical mixing* or *beating methods* are usually used for processes that occur on time scales slower than about 10^{-6} sec.

4 · 2 FILTER TECHNIQUES

The filter method involves the spectral decomposition of the scattered light by a diffraction grating or a Fabry-Perot interferometer. These devices are interposed between the scattering sample and a photomultiplier tube and act as filters, which for a given setting pass only a single frequency component of the scattered light. The filter output is then incident upon a photomultiplier cathode (PM) whose average dc output is proportional to the spectral density of the scattered electric field at the filter frequency. The filter is then swept through a range of frequencies. As discussed in Section 2.4, the PM output is proportional to the spectral density of the scattered electric field E_s

$$I_E(\omega_f) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt \exp i\omega_f t \langle E_s^*(0) E_s(t) \rangle \quad (4.2.1)$$

where ω_f is the frequency of the scattered light. Now we note from Eq. (3.2.8) that $E_s(t)$ contains an amplitude factor $\delta\epsilon_{if}(\mathbf{q}, t)$ and a phase factor $\exp i(k_f R - \omega_f t)$. Substitution of Eq. (3.2.8) into Eq. (4.2.1) then shows that the output of the filter is proportional to the spectral density $I_{if}^s(\mathbf{q}, \omega)$ or $I_{if}^a(\mathbf{q}, \omega)$ where $\omega \equiv \omega_i - \omega_f$ is the frequency shift of the scattered light [see Eqs. (3.3.3) and (3.2.12)]. Thus the quantity directly measured in filter experiments is proportional to the Fourier transform of the dielectric or polarizability time-correlation functions $I_{if}^s(\mathbf{q}, t)$ or $I_{if}^a(\mathbf{q}, t)$ given, respectively, by Eqs. (3.2.15) and (3.3.3b). In the ensuing chapters we often let $I_{if}(\mathbf{q}, t)$ [or $I_{if}^{(1)}(\mathbf{q}, t)$] represent either $I_{if}^s(\mathbf{q}, t)$ or $I_{if}^a(\mathbf{q}, t)$.

For processes faster than about 10^{-10} sec, diffraction gratings are used as the filter. For the slower processes with relaxation times in the range of about 10^{-6} – 10^{-10} sec Fabry-Perot interferometers are used (See Appendix 4.A). Filter experiments do not have sufficient resolution to study processes slower than about 10^{-6} sec. For fluctuations in the time range 1 – 10^{-6} sec optical mixing techniques must be used.

4 · 3 OPTICAL MIXING TECHNIQUES

Optical mixing techniques are the optical analogs of the beating techniques developed in radio-frequency spectroscopy (Forrester, 1961). They have made possible the application of light scattering to the study of the dynamics of relatively slow processes such as macromolecular diffusion, the dynamics of fluctuations in the critical region, and the motility of microorganisms.

In optical mixing methods no "filter" is inserted between the scattering medium and the photomultiplier (see Fig. 1.2.2). The scattered light impinges directly on the PM cathode. In the *homodyne* (or self-beat) method only the scattered light impinges on the photocathode, while in the *heterodyne* method a *local oscillator* (usually a small portion of the unscattered laser beam) is mixed with the scattered light on the cathode surface. Since the phototube is a square-law detector, its instantaneous current output is proportional to the square of the incident electric field $i(t) \propto |E(t)|^2$. The square of the electric field is proportional to the intensity of the light (or in quantum language, the number of photons).

The PM output is then usually passed into a hardwire computer called an *autocorrelator*, which calculates its time autocorrelation function

$$\langle i(t) i(0) \rangle = B \langle |E(0)|^2 |E(t)|^2 \rangle \quad (4.3.1)$$

where B is a proportionality constant.¹ The autocorrelator can be used in either a "digital" or an "analog" mode (see Appendix 4.B). In the digital method one counts and then autocorrelates current pulses (or photons) and in the analog method, one directly autocorrelates the fluctuations in the PM output current. For low light levels the digital method is normally preferred. (Jakeman and Pike, 1968; 1969).

For the purpose of discussing the differences between heterodyne and homodyne scattering we define the two scattered field autocorrelation functions

$$I_1(t) \equiv \langle E_s^*(0) E_s(t) \rangle \quad (4.3.2a)$$

$$I_2(t) \equiv \langle |E_s(0)|^2 |E_s(t)|^2 \rangle \quad (4.3.2b)$$

Homodyne Method

Since in the homodyne method only the scattered light impinges on the photocathode, $E(t)$ in Eq. (4.3.1) is equal to the scattered field $E_s(t)$, so that $\langle i(0)i(t) \rangle$ is proportional to $I_2(t)$ —which is consequently sometimes called the *homodyne correlation function*. The amplitude of $E_s(t)$, the scattered field, is proportional to the instantaneous dielectric constant fluctuations in the scattering volume and, of course, fluctuates in the same manner. In certain circumstances the homodyne correlation function may be simply expressed in terms of $I_1(t)$ or equivalently $I_{ij}^s(\mathbf{q}, t)$ or $I_{ij}^p(\mathbf{q}, t)$ of Eqs. (3.2.15) and (3.3.3b), respectively, as we now discuss.

The scattering volume V can be subdivided into subregions of volume small compared to the wavelength of light. Then the scattered field E_s can be regarded as a superposition of fields from each of the subregions, so that

$$E_s = \sum_n E_s^{(n)}$$

where $E_s^{(n)}$ is the scattered field from the n th subregion. As particles move $E_s^{(n)}$ fluctuates. If, as is often the case, the subregions are sufficiently large to permit particle motions in one subregion to be independent of those in another, E_s can be regarded as a sum of independent random variables ($E_s^{(1)}, E_s^{(2)}, \dots$). In this eventuality, the central limit theorem implies that E_s , which is itself a random variable, must be distributed according to a Gaussian distribution. A Gaussian distribution is completely characterized by its first and second moments. It follows that all higher moments of this distribution are related to the first two moments. From considerations of this kind (see Appendix 4.C) it is easy to show that the quantity in Eq. (4.3.2b), which is a fourth moment of the distribution, is related to $I_1(t)$, which is the second moment, through the equation.

$$I_2(t) = |I_1(0)|^2 + |I_1(t)|^2 \quad (4.3.3)$$

The important assumption in deriving this result is that the scattering volume can be divided into a large number of statistically independent subregions.² There are circumstances in which this assumption may be invalid. For example, systems in their critical region have very long correlation lengths; hence care must be exercised in applying Eq. (4.3.3) to critical fluids. In particular, the scattering volume must be chosen to contain a sufficiently large number of *correlation-volumes* to justify the use of the central limit theorem.

Another example involves the scattering from either dilute macromolecular solutions or bacterial dispersions. The scattered field is a superposition of the amplitudes scattered from each of these large independent particles. If the scattering volume V is sufficiently large, there will be on the average a large number $\langle N \rangle$ of particles in V and consequently a large number of terms in the superposition, thereby justifying the use of the central limit theorem to derive Eq. (4.3.3). However if either the concentration c or the volume is small so that $\langle N \rangle = cV$ is small, there might be an insufficient number of terms in the superposition so that the central limit theorem, and thereby Eq. (4.3.3), is invalid. In Section 5.5 we show that in this eventuality, I_2 contains more

information than I_1 and moreover can be used effectively to study additional properties of these solutions. Although these conditions are probably satisfied in the vast majority of applications, the restrictions must be kept in mind in any application of the homodyne technique.

Let us now return to Eq. (4.3.3). The $|I_1(0)|^2$ term is a dc term that determines the baseline for the homodyne correlation function of the scattered light.³ If, as is often the case, $I_1(t)$ is a sum of exponentials,

$$I_1(t) = \sum_i a_i \exp \frac{-t}{\tau_i} \quad (4.3.4)$$

then in the Gaussian approximation [Eq. (4.3.3)] the corresponding $I_2(t)$ is

$$I_2(t) = \sum_{i,j} a_i a_j \left[1 + \exp -t \left(\frac{1}{\tau_i} + \frac{1}{\tau_j} \right) \right] \quad (4.3.5)$$

If an autocorrelator is used to analyze the PM output the correlation function in Eq. (4.3.5) is directly measured. Sometimes a spectrum analyzer⁴ is used instead of an autocorrelator. This device determines the spectral density of the photocurrent from the PM and thus in a homodyne experiment it determines the time Fourier transform of $I_2(t)$, that is,

$$I_2(\omega) = \sum_{i,j} a_i a_j \left[\delta(\omega) + \frac{1}{\pi} \frac{\left(\frac{1}{\tau_i} + \frac{1}{\tau_j} \right)}{\left(\omega^2 + \left(\frac{1}{\tau_i} + \frac{1}{\tau_j} \right)^2 \right)} \right] \quad (4.3.6)$$

where the delta function arises from the constant term in Eq. (4.3.5) (which is equivalent to the term $|I_1(0)|^2$ in Eq. (4.3.3)).

Note from Eqs. (4.3.5) that more exponentials may appear in $I_2(t)$ than in $I_1(t)$. Similarly from Eq. (4.3.6), more Lorentzians may appear in $I_2(\omega)$ than in $I_1(\omega)$. For instance, if $I_1(t)$ is composed of two exponentials

$$I_1(t) = a_1 \exp \frac{-t}{\tau_1} + a_2 \exp \frac{-t}{\tau_2} \quad (4.3.7)$$

then $I_2(t)$ is composed of three exponentials

$$I_2(t) = (a_1 + a_2)^2 + a_1^2 \exp -\frac{2t}{\tau_1} + a_2^2 \exp -\frac{2t}{\tau_2} + 2a_1 a_2 \exp -t \left(\frac{1}{\tau_1} + \frac{1}{\tau_2} \right) \quad (4.3.8)$$

and $I_2(\omega)$ is composed of three Lorentzians

$$I_2(\omega) = (a_1 + a_2)^2 \delta(\omega) + \frac{1}{\pi} \left\{ a_1^2 \frac{\frac{2}{\tau_1}}{\omega^2 + \left(\frac{2}{\tau_1} \right)^2} + a_2^2 \frac{\frac{2}{\tau_2}}{\omega^2 + \left(\frac{2}{\tau_2} \right)^2} + 2a_1 a_2 \frac{\left(\frac{1}{\tau_1} + \frac{1}{\tau_2} \right)}{\omega^2 + \left(\frac{1}{\tau_1} + \frac{1}{\tau_2} \right)^2} \right\} \quad (4.3.9)$$

For processes with several relaxation times, the analysis of homodyne data is quite difficult. For a one-relaxation time process, however, the homodyne correlation function (or spectral density) is still exponential (Lorentzian), except that the relaxation time that appears is $\tau/2$ rather than τ .

Heterodyne Method

In the heterodyne method, a small portion of the unscattered laser light (local oscillator) is mixed with the scattered light on the photomultiplier cathode (see Fig. 1.2.2c). We assume below that the local oscillator varies at the laser frequency.² The PM output may then be analyzed with an autocorrelator or a spectrum analyzer.

If $E_{LO}(t)$ represents the local oscillator electric field, then the electric field at the PM is the superposition of $E_{LO}(t)$ and $E_s(t)$ and thus the autocorrelation function of the PM output, Eq. (4.3.1), becomes

$$\langle i(0)i(t) \rangle = B \langle |E_{LO}(t) + E_s(t)|^2 |E_{LO}(0) + E_s(0)|^2 \rangle \quad (4.3.10)$$

By proper choice of the experimental conditions, the amplitude of the local oscillator may be made much greater than the amplitude of the scattered field

$$|E_{LO}(t)| \gg |E_s(t)|$$

The following approximations are also made: (a) fluctuations of the local oscillator field are negligible and (b) the local oscillator field and the scattered field are statistically independent so that, for example, $\langle I_s I_{LO} \rangle = \langle I_{LO} \rangle \langle I_s \rangle$.

With these assumptions the expansion of Eq. (4.3.10) yields 16 terms, 10 of which are zero, three dc, and one negligible (the term $\langle |E_s(0)|^2 |E_s(t)|^2 \rangle$). The remaining two terms, the only ones of importance for our consideration, give

$$\langle i(0)i(t) \rangle \cong B [I_{LO}^2 + 2I_{LO} \text{Re} I_1(t)] \quad (4.3.11)$$

where

$$I_{LO} = \langle |E_{LO}|^2 \rangle$$

is the intensity of the local oscillator signal, and $\text{Re} I_1(t)$ is the real part⁶ of $I_1(t)$. Thus $\text{Re} I_1(t)$ is sometimes called the *heterodyne correlation function*. The background or dc term in the heterodyne experiment is determined by the LO intensity I_{LO} .

Note that the heterodyne technique introduces no extra terms into the scattered field time-correlation function, as contrasted to the homodyne spectrum of a many-exponential process.

Because $E_s(t)$ is proportional to $\delta \epsilon_{if}(\mathbf{q}, t)$, or correspondingly, $\delta \alpha_{if}(\mathbf{q}, t)$, $I_1(t)$ and $I_2(t)$ are, respectively, proportional to the correlation functions₂

$$I_{if}^{(1)}(\mathbf{q}, t) = \langle \delta \epsilon_{if}^*(\mathbf{q}, 0) \delta \epsilon_{if}(\mathbf{q}, t) \rangle \quad (4.3.12a)$$

$$I_{if}^{(2)}(\mathbf{q}, t) = \langle | \delta \epsilon_{if}(\mathbf{q}, 0) |^2 | \delta \epsilon_{if}(\mathbf{q}, t) |^2 \rangle \quad (4.3.12b)$$

with proportionality constants A and A^2 defined by Eq. (3.2.14).

From Eqs. (4.3.12) it is clear that the homodyne and heterodyne techniques measure

different correlation functions of the dielectric constant fluctuations. In the event that the Gaussian approximation applies these two correlation functions are related to each other by Eq. (4.3.3),

$$I_{if}^{(2)}(\mathbf{q}, t) = |I_{if}^{(1)}(\mathbf{q}, 0)|^2 + |I_{if}^{(1)}(\mathbf{q}, t)|^2 \quad (4.3.13)$$

Exceptions to Eq. (4.3.13) are discussed in Chapter 5.

It should be noted that if a heterodyne experiment is performed with a spectrum analyzer rather than an autocorrelator, the resulting spectrum is proportional to⁷

$$I_{if}^{(1)}(\mathbf{q}, \omega) \equiv \frac{1}{2\pi} \int_{-\infty}^{+\infty} dt e^{-i\omega t} \text{Re} I_{if}^{(1)}(\mathbf{q}, t) \quad (4.3.14)$$

We shall call this the *heterodyne spectrum*. Comparing Eq. (4.3.14) with $I_{if}^{(1)}(\mathbf{q}, \omega)$ the spectrum measured by filter methods [see Eq. (4.2.1)], we see that only when $I_{if}^{(1)}(\mathbf{q}, t)$ is a real function of the time do the two experiments give the same spectrum. When currents are present in a system the resulting spectra are somewhat different, as we see in Chapter 5.

APPENDIX 4.A FABRY-PEROT INTERFEROMETER

A Fabry-Perot interferometer consists of two plane dielectric mirrors held parallel to each other (see Fig. 4.A.1). The inner surfaces of the mirrors are highly reflecting ($\approx 98\%$). In light-scattering experiments, light usually enters the interferometer normal to the mirrors and is then reflected back and forth in the interferometer cavity. The condition for survival in the cavity of a wave of wavelength λ is that an integral number of half wavelengths must fit in the cavity, that is,

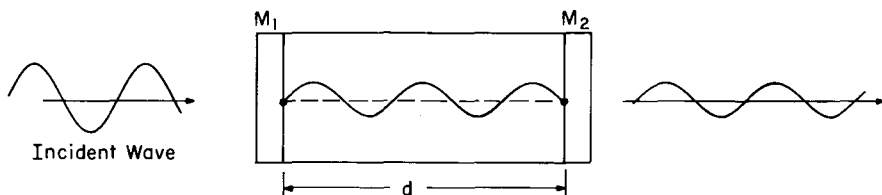


FIG. 4.A.1. Light enters the interferometer cavity normal to the mirror M . The wave is reflected back and forth in the cavity. Only waves for which a halfintegral number of wavelengths fit in the cavity survive. The mirrors are only partially reflecting so that the output wave contains only these allowed wavelengths.

$$m\left(\frac{\lambda}{2}\right) = d \quad (4.A.1)$$

where m is an integer and d the spacing between the mirrors. At this wavelength there will be a maximum in the intensity of the transmitted light (see Section 7.6.1 of Born and Wolf, 1964). If d is 0.1 cm and $\lambda = 5000 \text{ \AA}$, then $m = 2 \times 10^3$. The difference between successive wavelengths, $\Delta\lambda$, that can pass through the interferometer for this value of d is $\Delta\lambda \sim 2.5 \text{ \AA}$. The corresponding frequency range is $\Delta\nu \sim 3 \times 10^{11} \text{ Hz}$. This is large compared to the width of a typical Rayleigh-Brillouin line. Thus a proper

setting of d in the interferometer will select only one frequency component (wavelength) of the light-scattering spectrum. By varying d one can detect the spectral density of the scattered light as a function of frequency (see Section 4.2).

There are two common methods for sweeping the spectrum: (a) physically moving the plates piezoelectrically, thereby changing d and (b) changing λ (in the cavity) by changing the index of refraction n of the medium between the plates by, say, pumping a gas into the cavity.⁸ Moving the plates piezoelectrically has the disadvantage that it might upset the alignment of the plates. The index of refraction change method has the disadvantage of being time-consuming.

Some workers who want especially high resolution use spherical mirrors rather than the flat mirrors described above. Spherical mirrors increase the light-gathering power and also allow easier scanning of a spectrum.

There are two important parameters that characterize Fabry-Perot interferometers, the *free spectral range* (FSR) and the *finesse*. The FSR is essentially the frequency spacing between adjacent cavity modes. For a flat-plate interferometer, it is given by

$$\Delta\nu = \frac{c}{2nd}$$

and the finesse F is

$$F = \left(\frac{\Delta\nu}{\delta\nu} \right)$$

where $\delta\nu$ is the full width at half maximum of the instrumental linewidth of the interferometer. That there is any width at all is related to the fact that no mirror is perfectly reflecting or flat. The higher F is, the more clearly the interferometer distinguishes between different cavity modes. In addition, the impinging spectral linewidth should be small compared with the FSR otherwise the interferometer will give overlapping orders. Thus we desire an interferometer with large F and relatively large $\Delta\nu$.

APPENDIX 4.B OPTICAL MIXING EXPERIMENTS

A brief description of some of the important features of optical mixing experiments is given in this appendix. We include discussions of the concepts of "coherence" and "coherence area," their application to the calculation of predetection signal to noise ratios in optical mixing experiments, and a brief discussion of digital (photon counting) methods.

4 · B · 1 COHERENCE AREA

The *degree of coherence* of a light wave is essentially a measure of how close the wave is to a pure (or monochromatic) sine wave. Most waves are of limited duration and hence consist of many-frequency Fourier components. Such waves are not fully coherent. (They may be "partially" coherent.) Noncoherent waves have phase and amplitude

fluctuations that are random in space and time (Born and Wolf, 1964; Lipson and Lipson, 1969).

The concept of "coherence area" is central in considerations of the signal-to-noise ratio in optical mixing experiments. When light from an extended source impinges on a screen, a diffraction pattern is produced which depends among other things, on the extent of the source; that is, the intensity maxima and minima depend on the source dimensions. Figure 4.B.1 illustrates this for a one-dimensional case. Assume that each

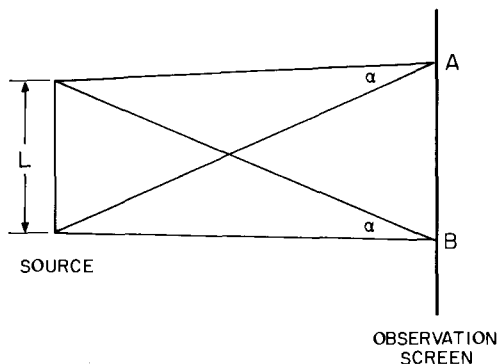


FIG. 4.B.1: A source of dimension L emits light that falls on the observation screen.

point on the source L radiates incoherently, that is, that light emitted from this point has random phase. Thus at point A on the screen the intensity will result from the superposition of waves arising from every point on the source. Consider now point B . If B is very close to A the signal at B would be almost identical to and "coherent" with that at A . In fact, when $A = B$, they are identical. How far apart must A and B be before this coherence disappears? This "coherence distance" is formally defined as the distance over which the spatial correlation function of the electric fields of the signals at A and B is significantly different from zero. In other words, it is the separation distance beyond which this correlation function $\langle E(A)E(B) \rangle$ has appreciably decayed.

A reasonable estimate of the coherence distance for the one-dimensional case shown in Fig. 4.B.1 may be obtained in the following way. First calculate the electric field at a point A on the screen. This field is the sum of those fields emanating from each point of the source. Thus

$$E(A) = \sum_i E(i)$$

where $E(i)$ is the field of the wave arriving at A from point i on the source. The field at B may be calculated in a similar manner,

$$E(B) = \sum_i E'(i)$$

where the field of the signal arriving at B from point i of the source $E'(i)$ differs from that arriving at A from the same point i by a phase factor which depends on the different path lengths traveled by the signal from point i to points A and B . It may then be shown by computing $\langle E(A)E(B) \rangle$ that the distance over which the signals at A and B are correlated (Lipson and Lipson, 1969) is roughly

$$l_c \approx \frac{\lambda}{\alpha} \quad (4.B.1a)$$

where α is the angle of the source subtended at the screen (see Fig. 4.B.1) and λ is the average wavelength of the radiation. If points A and B are separated by more than l_c the signals at A and B will not be appreciably correlated.

If the source were three-dimensional and the observation plane a flat surface such as the cathode of a photomultiplier tube, then one could define an *area* around the point A such that the signals at all points within this area are partially coherent with those at A . This area would be called the *coherence area*.

Estimates of the coherence area for typical light-scattering experiments have been given by several authors (Forrester, et al., 1955; Degiorgio and Lastovka, 1971). A useful estimate is

$$A_{\text{coh}} \approx \frac{\lambda^2}{\Omega} \quad (4.B.1b)$$

where Ω is the solid angle subtended by the source at the detector. Thus the smaller Ω , the larger A_{coh} .

4 · B · 2 PREDETECTION SIGNAL TO NOISE RATIOS IN ANALOG OPTICAL MIXING EXPERIMENTS

In an “*analog*” optical mixing experiment one measures either the time autocorrelation function of the photomultiplier output current or its corresponding spectral density. In the former case, the photomultiplier output is analyzed by an autocorrelator and in the latter, by a spectrum analyzer.

First let us consider what happens when the PM area A consists of

$$N = \frac{A}{A_{\text{coh}}}$$

coherence areas where A_{coh} is the coherence area.

The total PM current can be expressed as

$$i(t) = \sum_{j=1}^N i_j(t)$$

where $i_j(t)$ is the current from the j^{th} coherence area due to the scattered light. Furthermore, $i_j(t)$ can be expressed as

$$i_j(t) = \langle i_c \rangle + \delta i_j(t)$$

where $\langle i_c \rangle$ is the average current contribution of a coherence area and $\delta i_j(t)$ is the fluctuation of the current from the j^{th} coherence area. The autocorrelation function of the PM output is

$$\langle i(0)i(t) \rangle = \left\langle \sum_{j=1}^N \sum_{k=1}^N [\langle i_c \rangle + \delta i_j(0)] [\langle i_c \rangle + \delta i_k(t)] \right\rangle + \langle i_s(0)i_s(t) \rangle \quad (4.B.2a)$$

where in Eq. (4.B.2.a) we have added a shot-noise term $\langle i_s(0)i_s(t) \rangle$. The shot-noise term can be shown to be (Cummins and Swinney, 1970)

$$\langle i_s(0)i_s(t) \rangle = Ne \langle i_c \rangle \delta(t) \quad (4.B.2b)$$

where e is the electronic charge. Expanding the term in brackets in Eq. (4.B.2a) and noting that δi_j and δi_k are independent if $j \neq k$ (because by definition different coherence areas are uncorrelated), Eq. (4.B.2a) can be expressed as

$$\langle i(0)i(t) \rangle = N^2 \langle i_c \rangle^2 + N \langle \delta i_c(0) \delta i_c(t) \rangle + Ne \langle i_c \rangle \delta(t) \quad (4.B.2c)$$

where $\delta i_c(t)$ is the current fluctuation due to one typical coherence area.⁹

The term $\langle \delta i_c(0) \delta i_c(t) \rangle$ gives the time-dependence of the fluctuations measured by light scattering. In general, for a homodyne experiment

$$\frac{\langle \delta i_c(0) \delta i_c(t) \rangle}{\langle i_c \rangle^2} = \frac{I_2(t) - |I_1(0)|^2}{|I_1(0)|^2} \quad (4.B.2d)$$

where I_1 and I_2 are defined by Eqs. (4.3.2a) and (4.3.2b). In the Gaussian approximation Eq. (4.B.2d) reduces to

$$\frac{\langle \delta i_c(0) \delta i_c(t) \rangle}{\langle i_c \rangle^2} = \frac{|I_1(t)|^2}{|I_1(0)|^2} \quad (4.B.2e)$$

Defining $i_0 = N \langle i_c \rangle$, we can write Eq. (4.B.2c) in the Gaussian approximation as

$$\langle i(0)i(t) \rangle = i_0^2 \left[1 + \frac{1}{N} \frac{|I_1(t)|^2}{|I_1(0)|^2} \right] + ei_0 \delta(t) \quad (4.B.2f)$$

In a spectrum analyzer, the time Fourier transform $I(\omega)$ of Eq. (4.B.2f) is determined

$$I(\omega) = i_0^2 \left[\delta(\omega) + \frac{1}{2\pi N} \int_{-\infty}^{+\infty} dt \exp i\omega t \frac{|I_1(t)|^2}{|I_1(0)|^2} \right] + \frac{ei_0}{2\pi} \quad (4.B.2g)$$

Thus in spectrum analysis the shot noise contributes a constant (or dc) background $\frac{ei_0}{2\pi}$.

The predetection signal-to-noise ratio is the ratio of the coefficients of the integral containing $|I_1(t)|^2/|I_1(0)|^2$ to the shot-noise term, that is,

$$\frac{s}{n} = \frac{i_0}{Ne} = \frac{\langle i_c \rangle}{e} \quad (4.B.2h)$$

Thus for $N > 1$, s/n does not vary with N .

What happens when the illuminated area of the photocathode is less than one coherence area? In this case the fluctuating part of $\langle i(0)i(t) \rangle$ is proportional to i_0^2 (the

square of the total average dc photocurrent). Thus, the signal-to-noise ratio is now [cf. Eq. (4.B.2f)]

$$\frac{s}{n} = \frac{i_0}{e} = \frac{N\langle i_c \rangle}{e} \quad (4.B.2i)$$

Since i_0 increases linearly with the area A (or equivalently with $N = A/A_{coh}$), it may be seen from Eqs. (4.B.2h) and (4.B.2i) that s/n increases with A until $A = A_{coh}$ and then remains constant as A is increased beyond one coherence area. The important quantity in optical mixing experiments is therefore the signal per coherence area. It is advantageous in any given experiment to make the coherence area as large as possible. This may be done by making the scattering volume small, which is accomplished by focusing the laser beam to as small a volume as possible. This makes the solid angle Ω of Eq. (4.B.1b) small, thereby increasing A_{coh} .

Similar considerations apply to heterodyne spectroscopy. The predetection s/n ratio again depends on the signal per coherence area. In this case, however, the s/n also depends on the square of the heterodyning mixing efficiency. The heterodyne mixing efficiency represents the degree to which the scattered field and local oscillator field have matched phase fronts over an area equal to a coherence area (see, e.g., Benedek, 1969). The heterodyne mixing efficiency is unity if the local oscillator and the scattered light have wave fronts which match exactly in phase. If, however, the wave fronts are tilted or distorted relative to one another, the mixing efficiency will fall to a very small value. For strong scattered light signals, small values of the mixing efficiency may be tolerated, but for weak signals accurate matching must be achieved to obtain the high mixing efficiency necessary for signal processing. For the strong signal case, the local oscillator is usually taken to be unshifted laser light scattered from imperfections in the scattering cell.

The above considerations deal only with the predetection signal-to-noise ratio. The measuring device (spectrum analyzer or autocorrelator) will, for instance, introduce further errors. In addition, thermal noise in the electrical circuits has been ignored. In general, multichannel spectrum analyzers and autocorrelators give the best postdetection signal-to-noise ratios.

4 · B · 3 DIGITAL (PHOTOCOUNT) AUTOCORRELATION TECHNIQUES

In analog detection methods (See Section 4.B.2) the photomultiplier output is treated as a continuous variable. At low light-scattering levels, the signal-to-noise ratio in these methods may become small because of various sources of postdetection noise in the system (thermal or Johnson noise, PM dark current, etc.). Under these conditions it becomes advantageous to use the digital or photocount autocorrelation method.

In this technique one counts the number of photomultiplier output pulses in a given time and computes the time-autocorrelation function of these photocounts. Since each pulse, in the ideal case, corresponds to one arriving scattered photon, one measures the time-autocorrelation function of the number of photons arriving at the detector.

Thus if $n(t)$ is the number of photons arriving in the time interval between t and $t + \delta t$, the digital method gives

$$C(t) \equiv \langle n(t)n(0) \rangle \quad (4.B.3a)$$

The photocount rate is related to the instantaneous light intensity $I(R, t)$ which falls upon position R of the PM cathode at time t by the relation

$$n(t) = \sigma \int_A d^2R \int_t^{t+\delta t} dt' I(R, t') \quad (4.B.3b)$$

where σ is a constant related to the quantum efficiency of the photocathode and the spatial integration is over the illuminated area of the photocathode A . Although it is not explicitly indicated, $n(t)$ in Eq. (4.B.3b) depends on the sampling interval δt . From Eqs. (4.B.3a, b), the photocount correlation function is given by

$$C(t) = \sigma^2 \int_t^{t+\delta t} dt' \int_t^{t+\delta t} dt'' \int_A d^2R' \int_A d^2R'' \langle I(R', t') I(R'', t'') \rangle \quad (4.B.3c)$$

The integrations over the photocathode area A in Eq. (4.B.3c) depend on the coherence properties of the scattered light.

In the Gaussian approximation $C(t)$ for a photocount *homodyne*¹⁰ experiment reduces to (Jakeman and Pike, 1969)

$$C(t) = \langle n \rangle^2 \left[1 + f(A) \frac{|I_1(t)|^2}{|I_1(0)|^2} \right] \quad (4.B.3d)$$

where $\langle n \rangle \equiv \sigma(\delta t) \langle I \rangle$ is the average number of photocounts in the time interval δt , $I_1(t)$ is given by Eq. (4.3.2a) and $f(A)$ is a spatial coherence factor which depends on the number of coherence areas viewed and the sampling interval δt .

In practice $f(A)$ is usually determined by a fit to experimental data, whereas $\langle n \rangle$ is usually directly measured. In the case where $I_1(t)$ is a single exponential

$$I_1(t) = a \exp \frac{-t}{\tau} \quad (4.B.3e)$$

the photocount correlation function in the Gaussian approximation (Eq. (4.B.3d)) becomes

$$C(t) = \langle n \rangle^2 \left[1 + f(A) \exp \frac{-2t}{\tau} \right] \quad (4.B.3f)$$

Since $\langle n \rangle$ may be directly measured from the total count rate, only two parameters, $f(A)$ and τ , must be obtained from the experimental time-correlation function points. This is an advantage over the analog homodyne method in which, in the corresponding case, the measured $I_2(t)$ is fit to the form

$$I_2(t) = A + B \exp \frac{-2t}{\tau}$$

where A , B , and τ are parameters. Thus in the analog case the baseline A must in addition be obtained by a fit to the experimental time-correlation function points.

In measurements of the photocount correlation function an amplifier-discriminator system is placed after the PM. This system rejects small signals corresponding to noise in the circuits and amplifies and standardizes the PM pulses. The pulses are then counted and the photocount correlation function computed.

In order to attain fast sampling times and simplify the autocorrelator electronics, some workers do not use the "ideal" or "full" correlation methods described above, but instead use "clipping" and sometimes "scaling" methods.

A clipped correlator measures the correlation function $\langle n_k(0)n(t) \rangle$ where $n_k(t)$ is the clipped signal derived from the original $n(t)$ by

$$n_k(t) = \begin{cases} 1 & \text{when } n(t) > k \\ 0 & \text{when } n(t) \leq k \end{cases}$$

The integer k is called the *clipping level*. Thus instead of performing the multibit multiplication $n(0)n(t)$ and averaging, one has only to "multiply" $n(t)$ by 0 or 1. This "multiplication" may be achieved simply by using a "gate." That is, $n(t)$ is either passed or not passed into a storage register, depending on whether n_k is zero or one. Clipping simplifies the autocorrelator electronics considerably.

In the Gaussian approximation, the clipped autocorrelation function can be written in the form of Eq. (4.B.3d), where $f(A)$ now depends also on the clipping level, and the factor $\langle n \rangle^2$ is replaced by $\langle n_k \rangle \langle n \rangle$ (Jakeman and Pike, 1969). The clipping method has the disadvantage of being restricted to cases in which the Gaussian approximation applies. In cases in which the Gaussian approximation breaks down, (see Section 5.5) the "scaling" method is preferable (Koppel and Schaefer, 1973). In the scaling method one measures $\langle n^{(s)}_0(0)n(t) \rangle$, where $n^{(s)}_0 = 1$ if the s^{th} count occurs in the given time interval and zero otherwise. The scaling autocorrelation function is simply related to the intensity-autocorrelation function even for non-Gaussian scattered light if there is a sufficiently small count rate.

In conclusion let us note that there is an error due to the finite time average [cf. Eq. (2.4.7)] in the determination of the full autocorrelation functions. If T is the averaging time and τ_c is the correlation time, then there are effectively $(T/2\tau_c)$ "independent" samplings contributing to the correlation function. The relative error is proportional to the reciprocal of the square root of the number or $\pm(2\tau_c/T)^{1/2}$. A formula often used for an estimate of this error is

$$\Delta_T(t) = \pm \left(\frac{2\tau_c}{T} \right)^{1/2} [1 - C(t)]$$

where $C(t)$ is the normalized correlation function. Thus one must average over many correlation times in order to obtain reasonably small errors. This error is always present in addition to the other noise effects mentioned above.

APPENDIX 4.C THE GAUSSIAN APPROXIMATION

In general, the homodyne and heterodyne experiments yield different information about a system. Yet it is often asserted in the literature that precisely the same information is contained in both of these experiments. This assertion is based on the assumption that the complex property of Eq. (3.3.4) is a Gaussian stochastic variable. What

does this mean? Note that Eq. (3.3.4) involves a property of the form

$$\psi(\mathbf{q}, t) = \sum_{j=1}^N a_j \exp i\mathbf{q} \cdot \mathbf{r}_j(t),$$

which fluctuates as the particles in the system move around. If we know the joint probability that

$$\begin{aligned}\psi_0 &\leq \psi(\mathbf{q}, 0) \leq \psi_0 + d\psi_0 \\ \psi &\leq \psi(\mathbf{q}, t) \leq \psi + d\psi\end{aligned}$$

that is, if we know the joint probability that the fluctuating quantity $\psi(\mathbf{q}, t)$ has a value between ψ_0 and $\psi_0 + d\psi_0$ at time zero and between ψ and $\psi + d\psi$ at time t , then we can determine¹¹ the correlation functions $I_{if}^{(1)}$ and $I_{if}^{(2)}$ of Eq. (4.3.12),

$$\begin{aligned}I_{if}^{(1)}(\mathbf{q}, t) &= \int d\psi_0 \int d\psi \psi_0 P(\psi_0, \psi; t) \psi \\ I_{if}^{(2)}(\mathbf{q}, t) &= \int d\psi_0 \int d\psi |\psi_0|^2 P(\psi_0, \psi; t) |\psi|^2\end{aligned}$$

Here $P(\psi_0, \psi; t)$ is the joint probability distribution. If $\psi(\mathbf{q}, t)$ is a Gaussian stochastic variable, it can be shown (see, e.g., Wang and Uhlenbeck, 1945) that

$$\begin{aligned}P(\psi_0, \psi, t) &= [2\pi\langle|\psi|^2\rangle]^{-1} [1 - \hat{I}_{if}^{(1)}(\mathbf{q}, t)]^{-1/2} \exp -\psi_0^2/2\langle|\psi|^2\rangle \\ &\times \exp \left[\frac{(\psi - \psi_0 \hat{I}_1(\mathbf{q}, t))^2}{2\langle|\psi|^2\rangle[1 - \hat{I}_1^2(\mathbf{q}, t)]} \right]\end{aligned}$$

where

$$\hat{I}_1(\mathbf{q}, t) \equiv I_{if}^{(1)}(\mathbf{q}, t)/I_{if}^{(1)}(\mathbf{q}, 0)$$

If this expression is substituted into $I_{if}^{(2)}$ and the integration is carried out we obtain Eq. (4.3.2),

$$I_{if}^{(2)}(\mathbf{q}, t) = |I_{if}^{(1)}(\mathbf{q}, 0)|^2 + |I_{if}^{(1)}(\mathbf{q}, t)|^2$$

The assumption that ψ is a Gaussian random variable leads to the prediction that $|I_{if}^{(1)}|$ can be obtained from $I_{if}^{(2)}$. Homodyne detection then gives precisely the same information as heterodyne detection.

This Gaussian assumption is used so frequently in the interpretation of light scattering that it has become a traditional assumption. However it is not generally valid. We show this in the ensuing sections.

NOTES

1. B contains quantities related to the efficiency of the PM tube.
2. Relative deviations from Eq. (4.3.3) will be of order $1/n$ where n is the number of subregions.
3. This is the case even when the Gaussian approximation does not apply. For very long times

$$\langle |E_s(0)|^2 |E_s(t)|^2 \rangle \rightarrow \langle |E_s|^2 \rangle^2$$

4. In some sense a spectrum analyzer is equivalent to the insertion of an electronic filter between the PM and an averager.
5. Sometimes the unscattered light is modulated to change its frequency from the laser frequency.
6. In most applications $I_1(t)$ is a real even function of the time. It is not a real function, for example, for a system of charged particles in an external electric field or for any system in which there is a flow (cf. Section 5.8).
7. The full spectrum follows from Eqs. (4.3.11), (4.3.12a), and (3.2.14) and is

$$I'_{\text{f}}(\omega) = B[I_{\text{LO}}^2\delta(\omega) + 2AI_{\text{LO}}I_{\text{f}}^{(y)}(\omega)]$$

where we have omitted the contribution of the PM noise spectrum (see Appendix 4.B).

8. Since $\lambda = c/nv$, a change in n changes λ .
9. It is assumed that all coherence areas are equivalent.
10. The heterodyne case is discussed by Jakeman, 1970.
11. Let $\psi_0 = x_0 + iy_0$ and $\psi = x + iy$ where (x_0, y_0) and (x, y) are, respectively, the real and imaginary parts of ψ_0 and ψ . Then $d\psi = dx dy$ and $d\psi_0 = dx_0 dy_0$ are elements of area in the complex ψ and ψ_0 planes.

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